NON-ISOMORPHIC ENDOMORPHISMS OF FANO MANIFOLDS OF PICARD NUMBER ONE

JIHAO LIU

Abstract. We prove that a Fano manifold of Picard number one admitting a surjective endomorphism of degree greater than one is isomorphic to projective space, which confirms a conjecture that Hwang and Nakayama describe as folklore since the 1980s. As a consequence, a smooth projective rationally connected variety of Picard number one admitting an int-amplified endomorphism is isomorphic to projective space, and in particular toric, which answers the Picard number one case of questions of Fakhruddin and of Meng and Zhong.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

In [HN11], the following conjecture is described as folklore since the 1980s.

Conjecture 1.1 ([HN11, Conjecture 1.1])**.** *Let X be a Fano manifold of dimension d with $\rho(X) = 1$ and $X \ncong \mathbb{P}^d$. Then any surjective morphism $f : X \to X$ is bijective.*

Conjecture 1.1 is related to a question of Lazarsfeld [Laz84] (see [PS89, Problem 1]), which asks whether a smooth variety of dimension d that admits a finite surjective morphism of degree greater than one from a rational homogeneous space G/P, where P is a maximal parabolic subgroup of a semisimple algebraic group G, is isomorphic to $\mathbb{P}^d$. The question of Lazarsfeld was answered affirmatively in [HM99] (see also [HM04, Theorem 4]).

Amerik, following Van de Ven, conjectured that if X and Y are smooth projective varieties of dimension d with $b_2(X) = b_2(Y) = 1$, $Y \ncong \mathbb{P}^d$, and Y not an elliptic curve when $d = 1$, then the degree of a surjective morphism $X \to Y$ is bounded in terms of the discrete invariants of X and Y [Ame97, Conjecture B] (see also [ARV99, Conjecture 0.1]). For X in Conjecture 1.1, $H^2(X, \mathcal{O}_X) = 0$ by Kodaira vanishing [Har77, III, Remark 7.15], and hence $b_2(X) = \rho(X) = 1$. Thus [Ame97, Conjecture B] implies Conjecture 1.1. Indeed, [Ame97, Conjecture B] for the iterates f^n of a surjective morphism $f : X \to X$ bounds $(\deg f)^n = \deg(f^n)$ independently of n, which forces $\deg f = 1$, and then f is bijective (Lemma 2.11). [Bea01, Proposition 2(b)] shows that ramified endomorphisms occur only on varieties of Kodaira dimension $-\infty$, which leads

Date: October 26, 2026.

2020 *Mathematics Subject Classification.* 14J45, 14G17, 14M22, 08A35, 32H50.

Key words and phrases. Fano manifolds, Picard number one, non-isomorphic endomorphisms, Frobenius liftings, varieties of minimal rational tangents, toric varieties.

to the question of which Fano manifolds admit an endomorphism of degree greater than one. Fakhruddin asked whether a Fano manifold X of dimension d with $\mathrm{Pic}(X) \cong \mathbb{Z}$ that admits a surjective morphism $X \to X$ of degree greater than one is isomorphic to $\mathbb{P}^d$ [Fak03, Question 4.3].

Several known cases of Conjecture 1.1 concern minimal rational curves. Consider a family of rational curves on X such that the members through a general point $x \in X$ are parametrized by a non-empty proper scheme, for example, a family that covers X and has minimal degree among such families. For a general point $x \in X$, the tangent directions at x of the members through x form a subvariety of the projective space $\mathbb{P}(T_x X)$ of one-dimensional subspaces of $T_x X$, called the variety of minimal rational tangents (VMRT) of X. As in [HN11, Section 6], the family is fixed when we speak of the VMRT of X. The VMRT is linear if its irreducible components are linear subspaces of $\mathbb{P}(T_x X)$, which includes the case of a finite set [HN11, Section 6].

The *index* of X is the largest integer r such that $-K_X \sim rH$ for some Cartier divisor H on X, and X is a *del Pezzo manifold* (resp. a *Mukai manifold*) if the index of X is $d-1$ (resp. $d-2$). In each of the following cases, it is known that any surjective morphism $f : X \to X$ has degree one, which is equivalent to f being bijective (Lemma 2.11).

(1) When $d \leq 2$, by the classification of smooth projective curves and surfaces (see the remark after [Fak03, Question 4.3]).
(2) When $d = 3$ [Ame97, Theorem 2.3], [ARV99, Theorem 0.2 and Proposition 2.2], [HM03] (see [HN11, p. 1408]).
(3) When X is a rational homogeneous space [PS89, Proposition 2].
(4) When the identity component $\mathrm{Aut}^0(X)$ of the automorphism group of X has an open orbit in X [HN11, Theorem 1.4].
(5) When X is a smooth hypersurface in $\mathbb{P}^{d+1}$, of degree two [PS89, Proposition 8] or at least three [Bea01].
(6) When X is a non-linear smooth complete intersection in a projective space and $d \geq 3$ [SZ25, Proposition 1.9].
(7) When X is a smooth hypersurface of a Fano manifold, and X and the Fano manifold satisfy the conditions of [Cho11, Theorem 0.4].
(8) When $d = 4$ and the index of X is greater than one [SZ24, Theorem 1.5], [KT25, Theorem A].
(9) When X is a del Pezzo manifold [SZ24, Theorem 1.6] (for degree five, see also [AKP08, Theorem 7.21]).
(10) When X is a Mukai manifold [SZ25, Proposition 1.9].
(11) When the index of X is at most two and some line on X is not contained in the branch locus of f [AKP08, Theorem 7.6].
(12) When X belongs to one of the 32 currently known families of Fano fourfolds of index one [Wan26, Corollary 1.4].
(13) When X contains a rational curve with trivial normal bundle [HM03] (see [HN11, p. 1408]).
(14) When the VMRT of X is linear [HN11, Theorem 1.3].
(15) When T_X is big and the VMRT of X is not dual defective [SZ25, Theorem 1.2].
(16) When X is the moduli space of stable vector bundles of rank r with fixed determinant of degree coprime to r on a smooth projective curve of genus at least four [HR04, Theorem 5.6]. Such a moduli space is a Fano manifold with Picard group $\mathbb{Z}$ [HR04, p. 802].
(17) When f is étale outside a completely invariant divisor [HN11, Theorem 1.2].
(18) When f is Galois [JL26, Theorem 1.5].

We now explain recent approaches to Conjecture 1.1. [KT25, Theorem C] shows that a normal projective variety over a perfect field satisfies Bott vanishing for ample Weil divisors if it admits an int-amplified endomorphism (see Definition 2.10) whose degree is invertible in the base field. A Fano manifold satisfying Bott vanishing is rigid, and [KT25, Remark 1.3] deduces from [KT25, Theorem C] that only finitely many Fano manifolds in each dimension admit an

int-amplified endomorphism. Since $\rho(X) = 1$, any surjective endomorphism of X that is not an isomorphism is int-amplified (see [KT25, Remark 1.2]). Excluding a given X via [KT25, Theorem C] requires showing that Bott vanishing fails on X, and the known varieties have been treated family by family. For example, the three-dimensional case of [KT25, Theorem A] depends on the classification of Fano threefolds of Picard number one in characteristic zero (see [KT25, Section 1]), and [Wan26, Theorem 1.3, Corollary 1.4] excludes the 32 currently known families of Fano fourfolds of Picard number one and index one by computing $\chi(X, T_X) < 0$ for each of them. No such classification is known in general dimension.

Another approach uses Frobenius liftings. A smooth variety over an algebraically closed field k of characteristic $p > 0$ is F-liftable if its Frobenius morphism lifts to the ring $W_2(k)$ of Witt vectors of length two of k (see Definition 2.1). Achinger, Witaszek, and Zdanowicz conjectured that an F-liftable smooth projective variety X over k admits a finite étale Galois cover $Y \to X$ such that the Albanese morphism of Y is a toric fibration [AWZ21, Conjecture 1]. [AWZ21, Proposition 6.3.2] shows that an F-liftable smooth Fano variety X of dimension d over k with $\rho(X) = 1$ and T_X nef is isomorphic to $\mathbb{P}^d$. A surjective morphism $f : X \to X$ of a projective variety X is called *polarized* if there exist an ample Cartier divisor H on X and an integer $q > 1$ such that $f^*H \sim qH$. [AWZ21, Section 1.3] points out many similarities between Frobenius liftings and polarized endomorphisms.

Two steps are missing in order to deduce Conjecture 1.1 from [AWZ21, Proposition 6.3.2]. First, the analogy does not provide a way to obtain a Frobenius lifting of a reduction modulo p of X from a surjective endomorphism of X. Second, [AWZ21, Proposition 6.3.2] assumes that T_X is nef. When T_X is nef, the space of minimal free rational curves is proper [AWZ21, Lemma 6.1.3]. By a conjecture of Campana and Peternell [CP91] (see [AWZ21, Section 6, p. 2637]), a Fano manifold with nef tangent bundle should be rational homogeneous, and the rational homogeneous case of Conjecture 1.1 was settled by [PS89, Proposition 2]. [AWZ21, Section 6, p. 2637] expects the ideas there to be useful for the Picard number one case of [AWZ21, Conjecture 1] without the assumption that T_X is nef.

We now turn to minimal rational curves. [HN11, Theorem 1.3] is deduced from [HN11, Proposition 6.1] (based on [Hwa07, Propositions 2.1 and 2.2]), whose hypothesis is that the VMRT is linear and whose conclusion is that X has a web (see [HN11, Definition 5.1]), and [HN11, Theorem 5.8], which shows that any surjective endomorphism of a Fano manifold of Picard number one with a web is bijective. By [HN11, Theorem 1.3], Conjecture 1.1 reduces to the case where the VMRT of X is not linear, which is the common case. Indeed, as noted in [HN11, p. 1408], the only examples of Fano manifolds of Picard number one with linear VMRT known at the time contain rational curves with trivial normal bundle, and the VMRT of a rational homogeneous space of Picard number one other than projective space is not linear [HN11, p. 1425].

Hwang and Mok conjectured that for any family of rational curves on X for which the VMRT is defined, the VMRT at a general point is non-linear if its dimension is positive [HM04, Conjecture, p. 52]. [Hwa09, Theorem 1.3] shows that if the VMRT of X is non-linear or finite, then any surjective morphism $g : Y \to X$ from a projective variety Y satisfies $H^0(Y, g^*T_X) = g^*H^0(X, T_X)$, and in particular deformations of g come from $\mathrm{Aut}^0(X)$. Since [Hwa09, Theorem 1.3] concerns deformations of g, it does not exclude a surjective endomorphism of X of degree greater than one.

Another line of work concerns the positivity of T_X. [MZ26, Theorem 1.1] shows that the tangent sheaf of a normal compact Kähler variety with an int-amplified endomorphism is weakly positively curved. In particular, T_X is weakly positively curved if X has a surjective endomorphism that is not an isomorphism. For such X, Matsumura and Zhong conjectured that $H^0(X, \mathrm{Sym}^m T_X) \neq 0$ for some positive integer m, and more optimistically that T_X is big [MZ26, Conjecture 1.8]. [SZ25, Theorem 1.2] assumes that T_X is big and that the VMRT of X is not dual defective. The second condition fails already for the six-dimensional del Pezzo manifold of degree five, which is homogeneous: its VMRT is $\mathbb{P}^1 \times \mathbb{P}^2 \subset \mathbb{P}^5$ and is dual defective [SZ25, Example 3.2].

The goal of this paper is to prove Conjecture 1.1.

Theorem 1.2. *Let X be a Fano manifold of dimension d with $\rho(X) = 1$. Assume that there exists a surjective morphism $f : X \to X$ with $\deg f > 1$. Then $X \cong \mathbb{P}^d$.*

Theorem 1.2 is equivalent to Conjecture 1.1 by Lemma 2.11.

Theorem 1.2 answers [Fak03, Question 4.3] affirmatively. Theorem 1.2 also implies the conjecture of Matsumura and Zhong recalled above [MZ26, Conjecture 1.8]. Indeed, for a Fano manifold X of dimension d with $\rho(X) = 1$, a surjective morphism $X \to X$ that is not an isomorphism has degree greater than one (Lemma 2.11), so $X \cong \mathbb{P}^d$ by Theorem 1.2 if such a morphism exists. By the dual of the Euler sequence [Har77, II, Theorem 8.13], $T_{\mathbb{P}^d}$ is a quotient of $\mathcal{O}_{\mathbb{P}^d}(1)^{\oplus(d+1)}$, so $H^0(\mathbb{P}^d, T_{\mathbb{P}^d}) \neq 0$. Moreover, $T_{\mathbb{P}^d}$ is big, as is the tangent bundle of any rational homogeneous space [SZ25, Example 3.1].

Recall that a normal variety X is *toric* if X contains an algebraic torus as a dense open subset and the action of the torus on itself by multiplication extends to an action on X. Fakhruddin asked whether a smooth projective rationally connected variety (see Definition 2.4) admitting a polarized endomorphism is toric [Fak03, Question 4.4], and Meng and Zhong asked whether a smooth projective rationally connected variety admitting an int-amplified endomorphism is toric [MZ23, Question 1.2]. Both questions have positive answers for surfaces [Nak02, Theorem 3], for Fano threefolds [MZZ22, Theorem 1.4], for Fano fourfolds admitting a conic bundle structure [JZ23, Theorem 1.4], for Fano fourfolds of index greater than one and del Pezzo manifolds [SZ24, Corollary 1.9], for rationally connected threefolds of Picard number two [CMZ26, Theorem 1.3], and when the endomorphism is Galois [JL26, Theorem 1.4]. In general, a smooth projective rationally connected variety admitting an int-amplified endomorphism is of Fano type [Yos21, Corollary 1.4]. We deduce the following from Theorem 1.2.

Corollary 1.3. *Let X be a smooth projective rationally connected variety of dimension d with $\rho(X) = 1$. Assume that there exists a surjective morphism $f : X \to X$ that is int-amplified. Then $X \cong \mathbb{P}^d$.*

Since $\mathbb{P}^d$ is toric and polarized endomorphisms are int-amplified, Corollary 1.3 answers [Fak03, Question 4.4] and [MZ23, Question 1.2] affirmatively for varieties of Picard number one.

We also prove the following result in positive characteristic. We refer to Definition 2.1 for liftings over the ring $W_2(k)$ of Witt vectors of length two, liftings of the absolute Frobenius morphism F_X, and F-liftable varieties.

Theorem 1.4. *Let k be an algebraically closed field of characteristic $p > 0$, X a smooth projective variety over k, $f : X \to X$ a finite surjective morphism, and $q > 1$ an integer. Assume that ω_X^{-1} is ample, $p \nmid \deg f$, $f^*\omega_X^{-1} \cong (\omega_X^{-1})^{\otimes q}$, and there exist a lifting $\tilde{X}$ of X over $W_2(k)$ and a lifting $\tilde{f} : \tilde{X} \to \tilde{X}$ of f. Then F_X lifts to $\tilde{X}$. In particular, X is F-liftable.*

Sketch of the proof. To prove Theorem 1.2, suppose that $X \ncong \mathbb{P}^d$. By [HN11, Theorem 1.3], the VMRT of X is non-linear (Lemma 2.12), so a general minimal rational curve $g : \mathbb{P}^1 \to X$ satisfies

$$g^*T_X \cong \mathcal{O}_{\mathbb{P}^1}(2) \oplus \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus(d-1-s)} \oplus \mathcal{O}_{\mathbb{P}^1}^{\oplus s} \quad \text{with} \quad 1 \leq s \leq d-2$$

(Lemma 2.9), and $\dim W_g = s$ for the subspace $W_g \subset \Omega^1_X(g(0))$ of Definition-Proposition 2.6. Two properties of W_g contradict each other.

(i) In characteristic zero, W_g varies with g for fixed general $g(0)$ [Hwa09, Proposition 2.1], so the morphism κ sending $g \in M$ to W_g has non-zero relative differential over X on a non-empty open subset $M^\circ \subset M$ (Definition-Proposition 5.4).

(ii) In characteristic $p > 0$, a Frobenius lifting induces at a general point x a bijective p-semilinear map of the cotangent space, which stabilizes W_g for free g with $g(0) = x$ (Definition-Proposition 4.2) and has finitely many s-dimensional stable subspaces (Lemma 4.1). Thus W_g is locally constant in reduced families of such g with $\dim W_g = s$ (Corollary 4.3).

In Theorem 1.4, through normalized traces and Serre duality, the iterates of f kill the obstruction class to lifting the Frobenius morphism, so the reductions of X at the closed points of a smooth $\mathbb{Z}$-algebra are F-liftable (Proposition 3.3). Construction-Proposition 6.1 carries M° to them, where (i) contradicts (ii). The proof constructs no isomorphism $X \to \mathbb{P}^d$. Theorem 1.4, (ii), and the passage of (i) to the reductions are new, and f enters only through Lemma 2.12 and Proposition 3.3.

Each new link replaces a step at which a known argument fails.

(a) Frobenius liftings from the endomorphism. The descent of Frobenius liftings along a finite flat morphism of degree prime to p [AWZ21, Theorem 3.3.6(a)(ii)] needs a Frobenius lifting of the domain, so for $f : X \to X$ it presupposes the conclusion. Instead, in Theorem 1.4, we use that the line bundle of the ramification divisor of $h := f^n$ is $\omega_X \otimes h^*\omega_X^{-1} \cong (\omega_X^{-1})^{\otimes(q^n-1)}$, which is arbitrarily positive for large n. The functoriality of the obstruction class $o_{\tilde{X}} \in H^1(X, F_X^*T_X)$ to lifting F_X to $\tilde{X}$ under $\tilde{f}^n$ (Definition-Lemma 2.2(2)) and the normalized trace of h (Definition-Lemma 3.1(1)) make $o_{\tilde{X}}$ a fixed point of an operator, which vanishes by Serre duality once

$$H^1(X, F_X^*T_X \otimes (\omega_X^{-1})^{\otimes(q^n-1)}) = 0$$

(Definition-Lemma 3.1(2)), and Serre vanishing implies the vanishing for $n \gg 0$, but not for $n = 1$. The argument shows that a lifting of F_X to $\tilde{X}$ exists, without relating it to $\tilde{f}$.

(b) Rigidity at one point. [AWZ21, proof of Proposition 6.3.2] glues the sections of $g^*\Omega_X^1$ fixed by the Frobenius lifting along the family of minimal free rational curves g, which is proper as T_X is nef [AWZ21, Lemma 6.1.3], whereas the families carried from characteristic zero to the reductions of X are not proper. Instead, we use the lifting at one point x: for any free g with $g(0) = x$, evaluating at 0 the p-semilinear automorphism of $H^0(\mathbb{P}^1, g^*\Omega_X^1)$ induced by the lifting (cf. [AWZ21, Lemma 6.2.1]) shows that W_g is stable under the induced map φ_x of $\Omega_X^1(x)$ (Definition-Proposition 4.2). By Lemma 4.1, only finitely many subspaces of dimension s are stable under φ_x, since $V' \to \varphi_x(V')$ is a self-map of the Grassmannian with zero differential. The p-semilinearity is essential: f moves points and acts linearly on cotangent spaces, and a linear bijection, such as the identity, can stabilize all subspaces.

(c) An open condition across characteristics. The proof of (i) does not carry over to the reductions of X. In the proof of [Hwa09, Proposition 2.1], the projectivizations of the subspaces W_g sweep out the dual variety of a component of the VMRT, and the passage from the variation of W_g to a non-zero differential in Definition-Proposition 5.4 uses generic smoothness. Projective duality and generic smoothness both fail in positive characteristic, already for a smooth conic in characteristic 2, whose tangent lines all pass through one point. We therefore carry to the reductions only the condition that $d_{M/X}\kappa$ is non-zero at a point, which is open and compatible with base change (Construction-Proposition 6.1(4)). The differential is taken relative to X because (ii) concerns the curves through one point.

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce Frobenius liftings, the map on differential forms induced by a Frobenius lifting, the subspaces W_g for free morphisms g, and varieties of minimal rational tangents, and collect basic facts on endomorphisms of Fano manifolds of Picard number one.

In Section 3 we prove Theorem 1.4 and deduce from it Proposition 3.3 on reductions modulo p. In Section 4 we prove property (ii) in the form of Corollary 4.3.

In Section 5 we express property (i) by an open condition: we construct the morphism κ sending $g \in M$ to W_g, and show that the relative differential $d_{M/X}\kappa$ is non-zero on a non-empty open subset of M (Definition-Proposition 5.4). In Section 6 we spread out the endomorphism of Theorem 1.2 and the constructions of Section 5 over a finitely generated $\mathbb{Z}$-algebra (Construction-Proposition 6.1), and prove Theorem 1.2 and Corollary 1.3.

Acknowledgements. The work was partially supported by the National Key R&D Program of China #2024YFA1014400. Jihao Liu would like to thank Bin Dong, Guoxiong Gao, Zeming Sun,

Bin Wu, and Leheng Chen for AI assistance. Jihao Liu would like to thank Ruochuan Liu and Gang Tian for consistent support and encouragement.

2. Preliminaries

The three subsections contain the three kinds of input used in the rest of the paper: Frobenius liftings, rational curves, and endomorphisms.

2.1. **Frobenius liftings.**

The following definition is taken from [AWZ21, Section 1.6] and [AWZ21, Section 3].

Definition 2.1. Let k be a perfect field of characteristic $p > 0$ and $W_2(k)$ the ring of Witt vectors of length two of k, that is, the flat $\mathbb{Z}/p^2\mathbb{Z}$-algebra $W_2(k)$ together with an isomorphism $W_2(k)/pW_2(k) \cong k$, which is unique up to unique isomorphism. For a scheme X over $\mathbb{F}_p$, we denote by $F_X : X \to X$ the absolute Frobenius morphism. Let X be a smooth variety over k. A *lifting* of X over $W_2(k)$ is a flat $W_2(k)$-scheme $\tilde{X}$ together with an isomorphism $\tilde{X} \times_{\operatorname{Spec} W_2(k)} \operatorname{Spec} k \cong X$. Let $\tilde{X}$ be a lifting of X over $W_2(k)$, $\tilde{X}'$ a lifting of a smooth variety X' over $W_2(k)$, and $h : X \to X'$ a morphism. A *lifting* of h is a morphism $\tilde{h} : \tilde{X} \to \tilde{X}'$ over $W_2(k)$ whose restriction to X is h. We say that F_X *lifts* to $\tilde{X}$ if there exists a morphism $\tilde{F}_X : \tilde{X} \to \tilde{X}$ whose restriction to X is F_X. In this case, we say that $(\tilde{X}, \tilde{F}_X)$ is a *Frobenius lifting* of X. We say that X is *F-liftable* if X admits a Frobenius lifting.

Definition-Lemma 2.2 ([AWZ21, Proposition 3.3.1(a) and Lemma 3.3.3(a)])**.** Let k be a perfect field of characteristic $p > 0$, X a smooth variety over k, and $\tilde{X}$ a lifting of X over $W_2(k)$. Let $F_X^* T_X := (F_X^* \Omega_X^1)^\vee$, and let $o_{\tilde{X}} \in H^1(X, F_X^* T_X)$ be the obstruction class of [AWZ21, Proposition 3.3.1(a)] to lifting F_X to $\tilde{X}$. Then the following hold.

(1) F_X lifts to $\tilde{X}$ if and only if $o_{\tilde{X}} = 0$.
(2) Let X' be a smooth variety over k, $h : X \to X'$ a morphism, $\tilde{X}'$ a lifting of X' over $W_2(k)$, and $\tilde{h} : \tilde{X} \to \tilde{X}'$ a lifting of h. Then

$$h^* o_{\tilde{X}'} = H^1(F_X^*(dh))(o_{\tilde{X}}) \quad \text{in} \quad H^1(X, h^* F_{X'}^* T_{X'}),$$

where $dh : T_X \to h^* T_{X'}$ is the differential of h, and we identify $F_X^* h^* T_{X'}$ with $h^* F_{X'}^* T_{X'}$ by $h \circ F_X = F_{X'} \circ h$.

Proof. By [AWZ21, Proposition 3.3.1(a)], applied to liftings over $W_2(k)$, the obstruction class to lifting F_X to $\tilde{X}$ is an element of $\operatorname{Ext}^1(\Omega_X^1, F_{X*}\mathcal{O}_X)$, and F_X lifts to $\tilde{X}$ if and only if the obstruction class is zero. Since Ω_X^1 is locally free and F_X is affine, we have

$$\operatorname{Ext}^1(\Omega_X^1, F_{X*}\mathcal{O}_X) \cong H^1(X, T_X \otimes F_{X*}\mathcal{O}_X) \cong H^1(X, F_X^* T_X)$$

by [Har77, III, Propositions 6.3(c) and 6.7], the projection formula [Har77, II, Exercise 5.1(d)], and [Har77, III, Exercise 4.1]. Thus (1) holds. The equality in (2) is the left diagram in (3.3.1) of [AWZ21, Lemma 3.3.3(a)] (see also [Zda18, Lemma 4.1]), where the obstruction class is regarded as a morphism $F_X^* \Omega_X^1 \to \mathcal{O}_X[1]$, that is, an element of $H^1(X, F_X^* T_X)$. □

Definition-Lemma 2.3 ([AWZ21, Section 3.2 and Proposition 3.2.1(c)])**.** Let k be a perfect field of characteristic $p > 0$, X a smooth variety over k, and $(\tilde{X}, \tilde{F}_X)$ a Frobenius lifting of X. Then there exists an injective $\mathcal{O}_X$-linear morphism $\xi : F_X^* \Omega_X^1 \to \Omega_X^1$ such that $\xi(F_X^* dz) = z^{p-1} dz + dw$ for any local sections z and w of $\mathcal{O}_X$ with local lifts $\tilde{z}$ and $\tilde{w}$ satisfying $\tilde{F}_X^* \tilde{z} = \tilde{z}^p + p\tilde{w}$. In particular, the largest open subset $X^\circ \subset X$ over which ξ is an isomorphism is non-empty.

At any closed point x of X°, the map of fibers induced by the morphism ξ of Definition-Lemma 2.3, precomposed with the pullback by F_X, is a bijective p-semilinear map of the fiber of Ω_X^1 at x (see Lemma 4.1 and Definition-Proposition 4.2).

2.2. **Rational curves.** For the following definition, see [IP99, Definition-theorem 6.2.1].

Definition 2.4. Let X be a smooth projective variety. We say that X is *rationally connected* if for two general points $x_1, x_2 \in X$, there exists a rational curve on X containing x_1 and x_2.

Definition 2.5. Let X be a scheme and $x \in X$ a point. For an $\mathcal{O}_X$-module $\mathcal{F}$, we write $\mathcal{F}(x) := \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} k(x)$ for the fiber of $\mathcal{F}$ at x, and for a morphism ϕ of $\mathcal{O}_X$-modules, we write $\phi(x)$ for the induced map of fibers.

The notion of free morphisms is from [Kol96, II, Definition 3.1] (cf. [AWZ21, Definition 6.1.1]).

Definition-Proposition 2.6. Let k be an algebraically closed field, X a smooth variety over k, and $g : \mathbb{P}^1_k \to X$ a morphism. We say that g is *free* if $g^*T_X \cong \bigoplus_i \mathcal{O}_{\mathbb{P}^1}(a_i)$ with $a_i \geq 0$ for any i. Assume that g is free. Let $x := g(0)$ and let $W_g \subset \Omega^1_X(x)$ be the image of the evaluation map $H^0(\mathbb{P}^1_k, g^*\Omega^1_X) \to \Omega^1_X(x)$ at 0. Then the evaluation map is injective. In particular, $\dim W_g = h^0(\mathbb{P}^1_k, g^*\Omega^1_X)$.

Proof. Since g is free, $g^*\Omega^1_X \cong \bigoplus_i \mathcal{O}_{\mathbb{P}^1}(-a_i)$ with $a_i \geq 0$ for any i (see [Har77, V, Exercise 2.6] for the splitting of vector bundles on $\mathbb{P}^1_k$). The evaluation map at 0 is the direct sum of the evaluation maps at 0 of the summands $\mathcal{O}_{\mathbb{P}^1}(-a_i)$. If $a_i > 0$, then $H^0(\mathbb{P}^1_k, \mathcal{O}_{\mathbb{P}^1}(-a_i)) = 0$. If $a_i = 0$, then the evaluation map $H^0(\mathbb{P}^1_k, \mathcal{O}_{\mathbb{P}^1}) = k \to k$ at 0 is an isomorphism. Thus the evaluation map $H^0(\mathbb{P}^1_k, g^*\Omega^1_X) \to \Omega^1_X(x)$ at 0 is injective. □

Since g is free, W_g is a point of the Grassmannian of linear subspaces of $\Omega^1_X(x)$ of dimension $h^0(\mathbb{P}^1_k, g^*\Omega^1_X)$. The positive characteristic part and the characteristic zero part of the proof of Theorem 1.2 both study the subspaces W_g: under a Frobenius lifting, W_g does not vary in connected reduced families of free morphisms g with fixed $g(0)$ in X° and fixed $h^0(\mathbb{P}^1_k, g^*\Omega^1_X)$ (Section 4), while in characteristic zero, W_g varies with g (Section 5).

Definition 2.7. Let S be a scheme, $\mathcal{F}$ a locally free $\mathcal{O}_S$-module of finite rank n, and r an integer with $0 \leq r \leq n$. The *Grassmannian bundle* $\mathrm{Gr}_S(r, \mathcal{F}) \to S$ is the S-scheme such that for any S-scheme S', the S-morphisms $S' \to \mathrm{Gr}_S(r, \mathcal{F})$ correspond to the subsheaves of the pullback $\mathcal{F}_{S'}$ of $\mathcal{F}$ to S' that are locally free of rank r with locally free quotient. If $\mathcal{F}$ is free and $0 < r < n$, then $\mathrm{Gr}_S(r, \mathcal{F})$ is the Grassmannian over S of [Sta, Tag 089U], which is described there by the locally free quotients of rank $n - r$, and in general $\mathrm{Gr}_S(r, \mathcal{F})$ is obtained by gluing over an open cover of S on which $\mathcal{F}$ is free. If $r \in \{0, n\}$, then $\mathrm{Gr}_S(r, \mathcal{F}) = S$. For a field k and a k-vector space V with $\dim V < +\infty$, we write $\mathrm{Gr}(r, V) := \mathrm{Gr}_{\operatorname{Spec} k}(r, V)$. For a morphism $S' \to S$, we identify $\mathrm{Gr}_S(r, \mathcal{F}) \times_S S'$ with $\mathrm{Gr}_{S'}(r, \mathcal{F}_{S'})$ by the description of their S''-points for S'-schemes S''. In particular, the fiber of $\mathrm{Gr}_S(r, \mathcal{F})$ over a point $s \in S$ is $\mathrm{Gr}(r, \mathcal{F}(s))$.

Let Y be an S-scheme and $h : Y \to \mathrm{Gr}_S(r, \mathcal{F})$ a morphism over S. The *relative differential* of h over S is the natural map

$$d_{Y/S}h : h^*\Omega^1_{\mathrm{Gr}_S(r,\mathcal{F})/S} \to \Omega^1_{Y/S}$$

of [Har77, II, Proposition 8.11]. We say that $d_{Y/S}h$ is non-zero at a point $y \in Y$ if $(d_{Y/S}h)(y) \neq 0$. When $S = \operatorname{Spec} k$ for a field k, we write dh for $d_{Y/S}h$.

The following definition follows [HN11, Section 6] and [Hwa09, Section 2].

Definition 2.8. Let X be a Fano manifold. A *minimal component* of rational curves on X is an irreducible component $\mathcal{K}$ of the space $\mathrm{RatCurves}^n(X)$ of rational curves on X (see [Kol96, II, Definition-Proposition 2.11]) such that for a general point $x \in X$, the subscheme $\mathcal{K}_x \subset \mathcal{K}$ of members passing through x is non-empty and proper. For a general point $x \in X$, the *variety of minimal rational tangents* of $\mathcal{K}$ at x is the subvariety $\mathcal{C}_x \subset \mathbb{P}(T_xX)$ consisting of the tangent directions at x of members of $\mathcal{K}_x$, where $\mathbb{P}(T_xX)$ denotes the projective space of one-dimensional linear subspaces of T_xX. We also write VMRT for variety of minimal rational tangents. We say that the VMRT of $\mathcal{K}$ is *linear* if the irreducible components of $\mathcal{C}_x$ are linear subspaces of $\mathbb{P}(T_xX)$ for a general point $x \in X$. In particular, the VMRT of $\mathcal{K}$ is linear if $\mathcal{C}_x$ is a finite set for a general point $x \in X$. We say that the VMRT of $\mathcal{K}$ is *non-linear* if it is not linear.

As in [HN11, Section 6] and [Hwa09, Section 2], the condition on $\mathcal{K}$ in Definition 2.8 is that $\mathcal{K}_x$ is non-empty and proper for a general point x of X, not that the members of $\mathcal{K}$ have minimal $-K_X$-degree among the irreducible components of $\mathrm{RatCurves}^n(X)$ whose members cover X. The properness of $\mathcal{K}_x$ is the hypothesis of [Kol96, IV.2.9], which is used in Lemma 2.9. The convention that a finite $\mathcal{C}_x$ is linear is the one of [HN11, Section 6], under which [HN11, Theorem 1.3] holds.

Lemma 2.9. *Let X be a Fano manifold of dimension d with $\rho(X) = 1$ and $\mathcal{K}$ a minimal component of rational curves on X. Assume that the VMRT of $\mathcal{K}$ is non-linear. Then there exists an integer s with $1 \leq s \leq d-2$ such that the normalization $\nu : \mathbb{P}^1 \to X$ of a general member of $\mathcal{K}$ satisfies*

$$\nu^* T_X \cong \mathcal{O}_{\mathbb{P}^1}(2) \oplus \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus(d-1-s)} \oplus \mathcal{O}_{\mathbb{P}^1}^{\oplus s}.$$

In particular, $d \geq 3$.

Proof. By [Kol96, IV.2.9] and [Hwa09, Section 2], the normalization ν of a general member of $\mathcal{K}$ satisfies the isomorphism in the lemma for an integer s with $0 \leq s \leq d-1$, and the number $d-1-s$ of the summands $\mathcal{O}_{\mathbb{P}^1}(1)$ is equal to $\dim \mathcal{C}_x$ for a general point $x \in X$. If $s = d-1$, then $\dim \mathcal{C}_x = 0$, so $\mathcal{C}_x$ is a finite set for a general point $x \in X$, and the VMRT of $\mathcal{K}$ is linear. If $s = 0$, then $\dim \mathcal{C}_x = d-1 = \dim \mathbb{P}(T_x X)$, so an irreducible component of $\mathcal{C}_x$ is equal to $\mathbb{P}(T_x X)$ for a general point $x \in X$. Thus $\mathcal{C}_x = \mathbb{P}(T_x X)$ for a general point $x \in X$, and the VMRT of $\mathcal{K}$ is linear. Since the VMRT of $\mathcal{K}$ is non-linear, $1 \leq s \leq d-2$. In particular, $d \geq 3$. □

2.3. **Endomorphisms.** We recall the following definition from [Men20, Section 1].

Definition 2.10. Let X be a projective variety and $f : X \to X$ a surjective morphism. We say that f is *int-amplified* if $f^*H - H$ is ample for some ample Cartier divisor H on X.

Lemma 2.11. *Let X be a Fano manifold of dimension d with $\rho(X) = 1$ and $f : X \to X$ a surjective morphism. Then $\mathrm{Pic}(X)$ is generated by the class of an ample Cartier divisor H, and there exists a positive integer q such that $f^*H \sim qH$. Moreover, $\deg f = q^d$, f is finite, and $f^*\omega_X^{-1} \cong (\omega_X^{-1})^{\otimes q}$. In particular, $q \geq 2$ if and only if $\deg f > 1$. Furthermore, f is an isomorphism if $\deg f = 1$, and f is not bijective if $\deg f > 1$.*

Proof. By [IP99, Proposition 2.1.2(ii)(iii)], $\mathrm{Pic}(X)$ is torsion-free, and numerical equivalence coincides with linear equivalence on X. Since $\rho(X) = 1$, $\mathrm{Pic}(X)$ is generated by the class of a Cartier divisor H. Since $-K_X$ is ample, $-K_X \sim iH$ for some integer $i \neq 0$. Possibly replacing H with $-H$, we may assume that $i > 0$. Since $iH \sim -K_X$ is ample, H is ample [Har77, II, Proposition 7.5]. Write $f^*H \sim qH$ with $q \in \mathbb{Z}$. By the projection formula [Laz04, Remark 1.1.13],

$$q^d H^d = (f^*H)^d = \deg f \cdot H^d > 0.$$

Thus $q \neq 0$ and $\deg f = q^d$. Since f^*H is nef, $q > 0$. Hence f^*H is ample, so f does not contract any curve. Thus f is a projective morphism with finite fibers, and f is finite [Har77, III, Exercise 11.2]. Since $-K_X \sim iH$, we have $f^*\omega_X^{-1} \cong \mathcal{O}_X(iqH) \cong (\omega_X^{-1})^{\otimes q}$. Since $q \geq 1$ and $\deg f = q^d$, we have $q \geq 2$ if and only if $\deg f > 1$.

If $\deg f = 1$, then f is finite and birational, so $f_*\mathcal{O}_X$ is a coherent $\mathcal{O}_X$-algebra contained in the function field of X and integral over $\mathcal{O}_X$. As X is normal, $f_*\mathcal{O}_X = \mathcal{O}_X$. Since f is affine, f is an isomorphism. If $\deg f > 1$, then f is étale over a non-empty open subset of X [Har77, III, Corollary 10.7], so a general fiber of f consists of $\deg f$ points, and f is not bijective. □

Lemma 2.12. *Let X be a Fano manifold of dimension d with $\rho(X) = 1$ and $X \ncong \mathbb{P}^d$. Assume that there exists a surjective morphism $f : X \to X$ with $\deg f > 1$. Then there exists a minimal component of rational curves on X, and the VMRT of any minimal component of rational curves on X is non-linear.*

Proof. By [IP99, Corollary 6.1.5], X is covered by rational curves. Thus the members of some irreducible component of $\mathrm{RatCurves}^n(X)$ cover X, and we let $\mathcal{K}$ be such a component for which the $-K_X$-degree of the members is minimal among such components. Then $\mathcal{K}_x$ is non-empty

and proper for a general point $x \in X$ [Kol96, Chapter IV, Section 2], that is, $\mathcal{K}$ is a minimal component of rational curves on X.

Since $\deg f > 1$, f is not bijective by Lemma 2.11. Since $X \ncong \mathbb{P}^d$, the contrapositive of [HN11, Theorem 1.3] implies that the VMRT of any minimal component of rational curves on X is non-linear. □

3. Frobenius liftings from endomorphisms

In this section, we prove Theorem 1.4 and deduce Proposition 3.3, which yields Frobenius liftings on the reductions modulo p of families over rings smooth over $\mathbb{Z}$ that carry an endomorphism as in Theorem 1.4. The rigidity in Section 4 needs Frobenius liftings, and Section 6 uses Proposition 3.3 for one reduction.

To prove Theorem 1.4, we show that the obstruction class $o_{\tilde{X}}$ of the given lifting $\tilde{X}$ is fixed by an operator on $H^1(X, F_X^* T_X)$ built from an iterate of f and the normalized trace in Definition-Lemma 3.1, and that the operator vanishes for high iterates of f by Serre duality and Serre vanishing.

Definition-Lemma 3.1. Let k be an algebraically closed field, X a smooth projective variety of dimension d over k, and $h : X \to X$ a finite surjective morphism such that $\deg h$ is invertible in k. For a locally free sheaf $\mathcal{E}$ on X, let

$$\mathrm{tr}_{h,\mathcal{E}} : H^1(X, h^*\mathcal{E}) \to H^1(X, \mathcal{E})$$

be the map induced by the projection formula $h_* h^* \mathcal{E} \cong \mathcal{E} \otimes h_* \mathcal{O}_X$ and the normalized trace $(\deg h)^{-1} \mathrm{Tr}_h$ of the locally free $\mathcal{O}_X$-algebra $h_* \mathcal{O}_X$. Then the following hold.

(1) $\mathrm{tr}_{h,\mathcal{E}} \circ h^* = \mathrm{id}_{H^1(X,\mathcal{E})}$.

(2) Let $\beta : \mathcal{E} \to h^*\mathcal{E}$ be an $\mathcal{O}_X$-linear morphism. If $H^1(X, \mathcal{E} \otimes \omega_X \otimes h^* \omega_X^{-1}) = 0$, then $\mathrm{tr}_{h,\mathcal{E}} \circ H^1(\beta) = 0$.

Proof. Since X is smooth and h is finite and surjective, h is flat [Har77, III, Exercise 10.9]. Thus $h_* \mathcal{O}_X$ is a locally free $\mathcal{O}_X$-algebra of rank $\deg h$ [Har77, III, Proposition 9.2(e)], and the trace $\mathrm{Tr}_h : h_* \mathcal{O}_X \to \mathcal{O}_X$ is defined. Since h is affine, we identify $H^i(X, h^*\mathcal{E})$ with $H^i(X, h_* h^* \mathcal{E})$ for any i [Har77, III, Exercise 4.1]. Then $\mathrm{tr}_{h,\mathcal{E}}$ is induced by the composite

$$h_* h^* \mathcal{E} \cong \mathcal{E} \otimes h_* \mathcal{O}_X \xrightarrow{\mathrm{id}_{\mathcal{E}} \otimes (\deg h)^{-1} \mathrm{Tr}_h} \mathcal{E},$$

where the isomorphism is the projection formula [Har77, II, Exercise 5.1(d)].

We prove (1). The pullback $h^* : H^1(X, \mathcal{E}) \to H^1(X, h^*\mathcal{E})$ is induced by the morphism $\mathcal{E} \to h_* h^* \mathcal{E} \cong \mathcal{E} \otimes h_* \mathcal{O}_X$ sending a local section e to $e \otimes 1$. Since $\mathrm{Tr}_h(1) = \deg h$, the composite $\mathcal{E} \to \mathcal{E} \otimes h_* \mathcal{O}_X \to \mathcal{E}$ is $\mathrm{id}_{\mathcal{E}}$. Thus (1) holds.

We prove (2). If $d = 0$, then $H^1(X, \mathcal{E}) = 0$. Thus we may assume that $d \geq 1$. By duality for the finite flat morphism h [Har77, III, Exercise 6.10, Exercise 7.2(a), Proposition 7.2, Theorem 7.6(a), Corollary 7.12], there exists an isomorphism of $h_* \mathcal{O}_X$-modules

$$\mathcal{H}om_{\mathcal{O}_X}(h_* \mathcal{O}_X, \omega_X) \cong h_* \omega_X$$

such that, for any locally free sheaf $\mathcal{G}$ on X, Serre duality [Har77, III, Corollary 7.7] for $\mathcal{G}$ coincides with Serre duality for the locally free sheaf $h_* \mathcal{G}$, where we identify $H^i(X, \mathcal{G})$ with $H^i(X, h_* \mathcal{G})$, and $h_*(\mathcal{G}^\vee \otimes \omega_X)$ with $\mathcal{H}om_{\mathcal{O}_X}(h_* \mathcal{G}, \omega_X)$. Let $\phi : h^* \omega_X \to \omega_X$ be the morphism adjoint to the composite

$$\omega_X \to \mathcal{H}om_{\mathcal{O}_X}(h_* \mathcal{O}_X, \omega_X) \cong h_* \omega_X,$$

where the first morphism is the dual of $(\deg h)^{-1} \mathrm{Tr}_h$.

For $\mathcal{G} = h^* \mathcal{E}$, we have $h_* \mathcal{G} \cong \mathcal{E} \otimes h_* \mathcal{O}_X$, and hence Serre duality for $h^* \mathcal{E}$ is Serre duality for $\mathcal{E} \otimes h_* \mathcal{O}_X$. Moreover, $\mathrm{tr}_{h,\mathcal{E}}$ is induced by the morphism $\mathrm{id}_{\mathcal{E}} \otimes (\deg h)^{-1} \mathrm{Tr}_h : \mathcal{E} \otimes h_* \mathcal{O}_X \to \mathcal{E}$

of locally free sheaves. Since the isomorphisms of [Har77, III, Corollary 7.7] are natural, the transpose of $\mathrm{tr}_{h,\mathcal{E}}$ with respect to Serre duality is induced by the dual morphism

$$\mathcal{E}^\vee \otimes \omega_X \to \mathcal{E}^\vee \otimes \mathcal{H}om_{\mathcal{O}_X}(h_*\mathcal{O}_X, \omega_X) \cong \mathcal{E}^\vee \otimes h_*\omega_X \cong h_*(h^*\mathcal{E}^\vee \otimes \omega_X),$$

which is adjoint to $\mathrm{id}_{h^*\mathcal{E}^\vee} \otimes \phi$. Thus the transpose of $\mathrm{tr}_{h,\mathcal{E}}$ is the composite

$$H^{d-1}(X, \mathcal{E}^\vee \otimes \omega_X) \xrightarrow{h^*} H^{d-1}(X, h^*\mathcal{E}^\vee \otimes h^*\omega_X) \xrightarrow{H^{d-1}(\mathrm{id}_{h^*\mathcal{E}^\vee} \otimes \phi)} H^{d-1}(X, h^*\mathcal{E}^\vee \otimes \omega_X).$$

By the naturality in [Har77, III, Corollary 7.7] again, the transpose of $H^1(\beta)$ is $H^{d-1}(\beta^\vee \otimes \mathrm{id}_{\omega_X})$, where $\beta^\vee : h^*\mathcal{E}^\vee \to \mathcal{E}^\vee$ is the dual of β. Since

$$(\beta^\vee \otimes \mathrm{id}_{\omega_X}) \circ (\mathrm{id}_{h^*\mathcal{E}^\vee} \otimes \phi) = (\mathrm{id}_{\mathcal{E}^\vee} \otimes \phi) \circ (\beta^\vee \otimes \mathrm{id}_{h^*\omega_X}),$$

the transpose of $\mathrm{tr}_{h,\mathcal{E}} \circ H^1(\beta)$ is

$$H^{d-1}(\mathrm{id}_{\mathcal{E}^\vee} \otimes \phi) \circ H^{d-1}(\beta^\vee \otimes \mathrm{id}_{h^*\omega_X}) \circ h^*,$$

which factors through $H^{d-1}(X, \mathcal{E}^\vee \otimes h^*\omega_X)$. By Serre duality, $H^{d-1}(X, \mathcal{E}^\vee \otimes h^*\omega_X)$ is dual to $H^1(X, \mathcal{E} \otimes \omega_X \otimes h^*\omega_X^{-1})$, which is zero by assumption. Thus the transpose of $\mathrm{tr}_{h,\mathcal{E}} \circ H^1(\beta)$ is zero, and hence $\mathrm{tr}_{h,\mathcal{E}} \circ H^1(\beta) = 0$. □

Proof of Theorem 1.4. Let $o_{\tilde{X}} \in H^1(X, F_X^*T_X)$ be the obstruction class of Definition-Lemma 2.2 for the given lifting $\tilde{X}$, and let $\mathcal{E} := F_X^*T_X$, which is locally free since X is smooth. Let n be a positive integer, $h := f^n$, and $\tilde{h} := \tilde{f}^n$. Then h is finite and surjective, $\deg h = (\deg f)^n$ is invertible in k since $p \nmid \deg f$, and $\tilde{h}$ is a lifting of h. Since $\tilde{h}$ maps $\tilde{X}$ to itself, Definition-Lemma 2.2(2) for $X' = X$ and $\tilde{X}' = \tilde{X}$ implies that

$$h^*o_{\tilde{X}} = H^1(\beta_h)(o_{\tilde{X}}) \quad \text{in} \quad H^1(X, h^*\mathcal{E}),$$

where $\beta_h := F_X^*(dh) : \mathcal{E} \to h^*\mathcal{E}$, and we identify $F_X^*h^*T_X$ with $h^*F_X^*T_X$ by $h \circ F_X = F_X \circ h$. By Definition-Lemma 3.1(1), we have

$$o_{\tilde{X}} = \mathrm{tr}_{h,\mathcal{E}}(h^*o_{\tilde{X}}) = \mathrm{tr}_{h,\mathcal{E}}\left(H^1(\beta_h)(o_{\tilde{X}})\right).$$

Since $f^*\omega_X^{-1} \cong (\omega_X^{-1})^{\otimes q}$, induction on n shows that $h^*\omega_X^{-1} \cong (\omega_X^{-1})^{\otimes q^n}$. Thus

$$\mathcal{E} \otimes \omega_X \otimes h^*\omega_X^{-1} \cong F_X^*T_X \otimes (\omega_X^{-1})^{\otimes (q^n-1)}.$$

Since ω_X^{-1} is ample and $q > 1$, Serre vanishing [Har77, III, Theorem 5.2; II, Theorem 7.6] implies that $H^1(X, F_X^*T_X \otimes (\omega_X^{-1})^{\otimes (q^n-1)}) = 0$ for $n \gg 0$. For such n, Definition-Lemma 3.1(2) implies that $\mathrm{tr}_{h,\mathcal{E}} \circ H^1(\beta_h) = 0$. Thus $o_{\tilde{X}} = 0$, and F_X lifts to $\tilde{X}$ by Definition-Lemma 2.2(1). In particular, X is F-liftable. □

Definition 3.2. Let R be a ring, k an algebraically closed field, and $t : \operatorname{Spec} k \to \operatorname{Spec} R$ a morphism. For an R-scheme X_R and a morphism h_R of R-schemes, we denote the base changes of X_R and h_R along t by $X_t := X_R \times_{\operatorname{Spec} R} \operatorname{Spec} k$ and h_t.

Theorem 1.4 needs a lifting of (X, f) over $W_2(k)$ and a degree prime to p. For families over rings smooth over $\mathbb{Z}$, both come from base change.

Proposition 3.3. *Let R be a finitely generated $\mathbb{Z}$-algebra that is smooth over $\mathbb{Z}$, $q > 1$ an integer that is invertible in R, $X_R \to \operatorname{Spec} R$ a smooth projective morphism with geometrically connected fibers, $f_R : X_R \to X_R$ a finite morphism over $\operatorname{Spec} R$, k an algebraically closed field, and $t : \operatorname{Spec} k \to \operatorname{Spec} R$ a morphism whose image is a closed point. Assume that $\omega_{X_R/R}^{-1}$ is ample over $\operatorname{Spec} R$ and $f_R^*\omega_{X_R/R}^{-1} \cong (\omega_{X_R/R}^{-1})^{\otimes q}$. Then k has positive characteristic, and X_t is F-liftable.*

Proof. Since the residue field of $\operatorname{Spec} R$ at the image of t is finite [Gro66, Lemme 10.4.11.1(i)], k has characteristic $p > 0$, and q is invertible in k.

Since R is smooth over $\mathbb{Z}$ and the kernel $pW_2(k)$ of $W_2(k) \to k$ satisfies $(pW_2(k))^2 = 0$, there exists a morphism $\tilde{t} : \operatorname{Spec} W_2(k) \to \operatorname{Spec} R$ whose restriction to $\operatorname{Spec} k$ is t [Gro67,

Définition 17.1.1, Définition 17.3.1]. Let $\tilde{X} := X_R \times_{\operatorname{Spec} R} \operatorname{Spec} W_2(k)$ and $\tilde{f}$ be the base changes of X_R and f_R along $\tilde{t}$. Since X_R is flat over $\operatorname{Spec} R$, $\tilde{X}$ is flat over $W_2(k)$ [Har77, III, Proposition 9.2(b)]. Moreover, $\tilde{X} \times_{\operatorname{Spec} W_2(k)} \operatorname{Spec} k \cong X_t$, and the restriction of $\tilde{f}$ to X_t is f_t.

Since $X_R \to \operatorname{Spec} R$ is smooth and projective with geometrically connected fibers, X_t is a smooth projective variety over k [Gro67, Proposition 17.3.3(iii)]. The base change f_t of the finite morphism f_R is a finite endomorphism of X_t, hence surjective.

Let $g : X_t \to X_R$ be the projection. Then $\omega_{X_t} \cong g^*\omega_{X_R/R}$ [Har77, II, Proposition 8.10], and $f_R \circ g = g \circ f_t$. Thus

$$f_t^*\omega_{X_t}^{-1} \cong g^* f_R^* \omega_{X_R/R}^{-1} \cong (\omega_{X_t}^{-1})^{\otimes q}.$$

Since $(\omega_{X_R/R}^{-1})^{\otimes m}$ is very ample over $\operatorname{Spec} R$ for some positive integer m [Har77, II, Theorem 7.6] and very ampleness is preserved under base change, $(\omega_{X_t}^{-1})^{\otimes m}$ is very ample, so $\omega_{X_t}^{-1}$ is ample.

Let $d := \dim X_t$, and let c be the class of $\omega_{X_t}^{-1}$ in the Chow ring of X_t. Then $f_t^* c = qc$. By the intersection theory on non-singular projective varieties over k [Har77, Appendix A, Theorem 1.1, A2 and A4], we have

$$\deg f_t \cdot \deg(c^d) = \deg\left((f_t^* c)^d\right) = q^d \deg(c^d).$$

Since mc is the class of a hyperplane section of X_t in the embedding defined by $(\omega_{X_t}^{-1})^{\otimes m}$, we have $\deg(c^d) > 0$. Thus $\deg f_t = q^d$, which is prime to p since q is invertible in k.

Therefore, $\tilde{X}$ is a lifting of X_t over $W_2(k)$, and $\tilde{f}$ is a lifting of f_t (Definition 2.1). By Theorem 1.4 for X_t, f_t, and q, X_t is F-liftable. □

4. Free rational curves on F-liftable varieties

In this section, we prove that for a smooth variety X with a Frobenius lifting and a closed point $x \in X^\circ$, only finitely many subspaces of $\Omega^1_X(x)$ are of the form W_g for a free morphism $g : \mathbb{P}^1_k \to X$ with $g(0) = x$, and hence W_g is locally constant in reduced families of such morphisms along which $\dim W_g$ is fixed (Corollary 4.3). In Section 6, we use Corollary 4.3 for reductions modulo p in the proof of Theorem 1.2.

The finiteness in Lemma 4.1 fails for k-linear bijections: any linear subspace is stable under the identity map.

Lemma 4.1. *Let k be an algebraically closed field of characteristic $p > 0$, V a k-vector space with $\dim V < +\infty$, and $\varphi : V \to V$ a bijective p-semilinear map, that is, φ is additive and $\varphi(cv) = c^p\varphi(v)$ for any $c \in k$ and $v \in V$. Then for any integer $s \geq 0$, there exist only finitely many linear subspaces $V' \subset V$ such that $\dim V' = s$ and $\varphi(V') = V'$.*

Proof. If $s > \dim V$, then no linear subspace of V is of dimension s. If $s = 0$ (resp. $s = \dim V$), then 0 (resp. V) is the only linear subspace of V of dimension s. Thus we may assume that $0 < s < \dim V$.

Let $V' \subset V$ be a linear subspace and $u_1, \dots, u_m$ a basis of V'. Since k is perfect and $\varphi(\sum_i b_i u_i) = \sum_i b_i^p \varphi(u_i)$ for any $b_1, \dots, b_m \in k$, $\varphi(V')$ is the linear span of $\varphi(u_1), \dots, \varphi(u_m)$. Since φ is injective and $\sum_i b_i \varphi(u_i) = \varphi(\sum_i b_i^{1/p} u_i)$ for any $b_1, \dots, b_m \in k$, $\varphi(u_1), \dots, \varphi(u_m)$ are linearly independent. Thus $\varphi(V')$ is a linear subspace of V with basis $\varphi(u_1), \dots, \varphi(u_m)$.

Let $n := \dim V$ and fix a basis $e_1, \dots, e_n$ of V. For $v \in V$, let $v_1, \dots, v_n \in k$ be the coordinates of v, and let $c_{ij} \in k$ be the i-th coordinate of $\varphi(e_j)$. Then the i-th coordinate of $\varphi(v)$ is

$$\varphi(v)_i = \sum_{j=1}^{n} c_{ij} v_j^p.$$

Since $\varphi(e_1), \dots, \varphi(e_n)$ are linearly independent, the matrix (c_{ij}) is invertible. Thus for any $v \in V$, $\varphi(v)$ is the image of the vector with coordinates $v_1^p, \dots, v_n^p$ under the linear automorphism of V with matrix (c_{ij}).

Let $\mathrm{Gr}(s,V)$ be the Grassmannian of Definition 2.7, whose closed points correspond to the linear subspaces of V of dimension s. Let $V'\subset V$ be a linear subspace with $\dim V'=s$, $u_1,\dots,u_s$ a basis of V', and $a_{lj}\in k$ the j-th coordinate of u_l. By the Plücker embedding, $\mathrm{Gr}(s,V)$ is a closed subvariety of $\mathbb{P}^N_k$, where $N=\binom{n}{s}-1$, and the Plücker coordinates of V' are the $s\times s$ minors of the matrix (a_{lj}). Let h be the morphism of $\mathbb{P}^N_k$ raising the Plücker coordinates to the p-th power. Since $c\to c^p$ is a ring endomorphism of k and the determinant is a polynomial in the entries with integer coefficients, the $s\times s$ minors of (a_{lj}^p) are the p-th powers of the $s\times s$ minors of (a_{lj}). Thus h maps V' to the linear subspace of V spanned by the vectors with coordinates $a_{l1}^p,\dots,a_{ln}^p$ for $1\le l\le s$. In particular, h maps the closed points of $\mathrm{Gr}(s,V)$ into $\mathrm{Gr}(s,V)$. Since $\mathrm{Gr}(s,V)$ is reduced and the closed points of $\mathrm{Gr}(s,V)$ are dense, h maps $\mathrm{Gr}(s,V)$ into $\mathrm{Gr}(s,V)$ [Har77, II, Exercises 3.11(d) and 3.14]. Let $f:\mathrm{Gr}(s,V)\to\mathrm{Gr}(s,V)$ be the composition of $h|_{\mathrm{Gr}(s,V)}$ and the automorphism of $\mathrm{Gr}(s,V)$ induced by the linear automorphism of V with matrix (c_{ij}). Then $f(V')$ is the linear span of $\varphi(u_1),\dots,\varphi(u_s)$, that is, $f(V')=\varphi(V')$.

Since $\Omega^1_{\mathrm{Gr}(s,V)}$ is locally generated by the differentials of the ratios z of two Plücker coordinates, $h^*z=z^p$, and $d(z^p)=pz^{p-1}dz=0$, we have $df=0$. Let y be a closed point of the fixed-point scheme of f, that is, of the intersection of the graph of f with the diagonal of $\mathrm{Gr}(s,V)\times_k\mathrm{Gr}(s,V)$. By [Har77, II, Exercise 2.8], the Zariski tangent space of the fixed-point scheme of f at y is $\ker(\mathrm{id}_{T_y\,\mathrm{Gr}(s,V)}-df_y)$, where $T_y\,\mathrm{Gr}(s,V)$ is the Zariski tangent space of $\mathrm{Gr}(s,V)$ at y and df_y is the tangent map of f at y. Since $df=0$, we have $df_y=0$ [Har77, II, Proposition 8.7], so $\ker(\mathrm{id}_{T_y\,\mathrm{Gr}(s,V)}-df_y)=0$. By Nakayama's lemma, the fixed-point scheme of f is reduced and of dimension zero, and hence consists of finitely many closed points. Since $f(V')=\varphi(V')$ for any linear subspace $V'\subset V$ with $\dim V'=s$, the closed points of the fixed-point scheme of f are the linear subspaces $V'\subset V$ with $\dim V'=s$ and $\varphi(V')=V'$. Thus there exist only finitely many such linear subspaces V'. □

To use Lemma 4.1, we attach to any closed point x of X° a bijective p-semilinear map φ_x of $\Omega^1_X(x)$ induced by the morphism ξ of Definition-Lemma 2.3, and show that φ_x preserves W_g for any free morphism g sending 0 to x.

Definition-Proposition 4.2. Let k be an algebraically closed field of characteristic $p>0$, X a smooth variety over k, and $(\tilde X,\tilde F_X)$ a Frobenius lifting of X. Let ξ and X° be as in Definition-Lemma 2.3. For a closed point $x\in X^\circ$, let $\varphi_x:\Omega^1_X(x)\to\Omega^1_X(x)$ be the map defined by $\varphi_x(w):=\xi(x)(F_X^*w)$, where F_X^*w denotes the pullback of w to the fiber of $F_X^*\Omega^1_X$ at x. Then φ_x is a bijective p-semilinear map. Moreover, for any free morphism $g:\mathbb{P}^1_k\to X$ with $g(0)=x$, we have $\varphi_x(W_g)=W_g$.

Proof. Since F_X is the identity on the underlying topological space of X and is the p-th power map on $\mathcal{O}_X$, the fiber of $F_X^*\Omega^1_X$ at x is $\Omega^1_X(x)\otimes_k k$, where the second factor k is a k-algebra through $c\to c^p$, and $F_X^*w=w\otimes 1$ for any $w\in\Omega^1_X(x)$. Since k is perfect, $c\to c^p$ is an automorphism of k, and hence $w\to F_X^*w$ is a bijective p-semilinear map from $\Omega^1_X(x)$ to the fiber of $F_X^*\Omega^1_X$ at x. Since $x\in X^\circ$, $\xi(x)$ is bijective. Thus φ_x is a bijective p-semilinear map.

Let $g:\mathbb{P}^1_k\to X$ be a free morphism with $g(0)=x$. Since $F_X\circ g=g\circ F_{\mathbb{P}^1_k}$, the pullback of ξ by g is a morphism

$$g^*\xi:F_{\mathbb{P}^1_k}^*g^*\Omega^1_X\to g^*\Omega^1_X$$

whose fiber at 0 is $\xi(x)$. Since $\xi(x)$ is bijective and both $F_{\mathbb{P}^1_k}^*g^*\Omega^1_X$ and $g^*\Omega^1_X$ are locally free of rank $\dim X$, $g^*\xi$ is an isomorphism over a neighborhood of 0. Since $\mathbb{P}^1_k$ is integral and $F_{\mathbb{P}^1_k}^*g^*\Omega^1_X$ is locally free, $g^*\xi$ is injective.

Since g is free, $g^*\Omega^1_X\cong\bigoplus_i\mathcal{O}_{\mathbb{P}^1}(-a_i)$ with $a_i\ge 0$ for any i (Definition-Proposition 2.6). Since $F_{\mathbb{P}^1_k}^*$ raises the transition functions to the p-th power, $F_{\mathbb{P}^1_k}^*\mathcal{O}_{\mathbb{P}^1}(m)\cong\mathcal{O}_{\mathbb{P}^1}(pm)$ for any integer m,

and hence

$$F^*_{\mathbb{P}^1_k} g^* \Omega^1_X \cong \bigoplus_i \mathcal{O}_{\mathbb{P}^1}(-pa_i). \tag{4.1}$$

Thus $h^0(\mathbb{P}^1_k, g^*\Omega^1_X)$ and $h^0(\mathbb{P}^1_k, F^*_{\mathbb{P}^1_k} g^*\Omega^1_X)$ are both equal to the number of i with $a_i = 0$ [Har77, III, Theorem 5.1]. Since $g^*\xi$ is injective, the map

$$H^0(\mathbb{P}^1_k, F^*_{\mathbb{P}^1_k} g^*\Omega^1_X) \to H^0(\mathbb{P}^1_k, g^*\Omega^1_X)$$

induced by $g^*\xi$ is injective, and hence bijective. Under the decompositions $g^*\Omega^1_X \cong \bigoplus_i \mathcal{O}_{\mathbb{P}^1}(-a_i)$ and (4.1), the Frobenius pullback $\sigma \to F^*_{\mathbb{P}^1_k}\sigma$ from $H^0(\mathbb{P}^1_k, g^*\Omega^1_X)$ to $H^0(\mathbb{P}^1_k, F^*_{\mathbb{P}^1_k} g^*\Omega^1_X)$ raises the constant sections of the summands with $a_i = 0$ to the p-th power. Since k is perfect, the Frobenius pullback is a bijective p-semilinear map. Thus $\sigma \to (g^*\xi)(F^*_{\mathbb{P}^1_k}\sigma)$ is a bijective p-semilinear map of $H^0(\mathbb{P}^1_k, g^*\Omega^1_X)$ (cf. [AWZ21, Lemma 6.2.1]).

For $\sigma \in H^0(\mathbb{P}^1_k, g^*\Omega^1_X)$, let $\sigma(0) \in \Omega^1_X(x)$ be the image of σ under the evaluation map at 0. Since the fiber of $g^*\xi$ at 0 is $\xi(x)$ and the image of $F^*_{\mathbb{P}^1_k}\sigma$ in the fiber of $F^*_{\mathbb{P}^1_k} g^*\Omega^1_X = g^* F^*_X \Omega^1_X$ at 0 is $F^*_X(\sigma(0))$, we have

$$\big((g^*\xi)(F^*_{\mathbb{P}^1_k}\sigma)\big)(0) = \xi(x)\big(F^*_X(\sigma(0))\big) = \varphi_x(\sigma(0)).$$

Since W_g is the image of the evaluation map at 0 (Definition-Proposition 2.6) and $\sigma \to (g^*\xi)(F^*_{\mathbb{P}^1_k}\sigma)$ is a bijection of $H^0(\mathbb{P}^1_k, g^*\Omega^1_X)$, we have $\varphi_x(W_g) = W_g$. □

Corollary 4.3. *Let k, X, $(\tilde{X}, \tilde{F}_X)$, and X° be as in Definition-Proposition 4.2, let $x \in X^\circ$ be a closed point, and let s be an integer with $0 \le s \le \dim X$. Let T be a reduced k-scheme of finite type, $g_T : \mathbb{P}^1_k \times T \to X$ a morphism, and $\gamma : T \to \mathrm{Gr}(s, \Omega^1_X(x))$ a morphism. For any closed point $\tau \in T$, let $g_\tau := g_T|_{\mathbb{P}^1_k \times \{\tau\}}$. Assume that for any closed point $\tau \in T$, we have $g_\tau(0) = x$, g_τ is free, $h^0(\mathbb{P}^1_k, g^*_\tau \Omega^1_X) = s$, and $\gamma(\tau) = W_{g_\tau}$. Then γ is constant on any connected component of T. In particular, if T is smooth, then $d\gamma = 0$.*

Proof. By Definition-Proposition 4.2 and Lemma 4.1, there exist only finitely many linear subspaces $V' \subset \Omega^1_X(x)$ with $\dim V' = s$ and $\varphi_x(V') = V'$. Let $y_1, \dots, y_m$ be the closed points of $\mathrm{Gr}(s, \Omega^1_X(x))$ corresponding to such linear subspaces V'. For any closed point $\tau \in T$, we have $\dim W_{g_\tau} = h^0(\mathbb{P}^1_k, g^*_\tau \Omega^1_X) = s$ by Definition-Proposition 2.6, and $\varphi_x(W_{g_\tau}) = W_{g_\tau}$ by Definition-Proposition 4.2. Thus $\gamma(\tau) = W_{g_\tau}$ is one of $y_1, \dots, y_m$ for any closed point $\tau \in T$, that is, the closed subset $\gamma^{-1}(\{y_1, \dots, y_m\})$ of T contains all closed points of T. Since T is of finite type over k, the closed points of T are dense in T [Har77, II, Exercise 3.14], and hence $\gamma^{-1}(\{y_1, \dots, y_m\}) = T$. Thus T is the disjoint union of the closed subsets $\gamma^{-1}(y_1), \dots, \gamma^{-1}(y_m)$, and hence $\gamma^{-1}(y_i)$ is open and closed in T for any i.

Let T' be a connected component of T. Then $T' \subset \gamma^{-1}(y_i)$ for some i. Since T' is reduced, $\gamma|_{T'}$ factors through the reduced closed subscheme $\{y_i\} \cong \operatorname{Spec} k$ of $\mathrm{Gr}(s, \Omega^1_X(x))$ [Har77, II, Exercise 3.11(d)], that is, γ is constant on T'. Since $d\gamma$ is the natural map $\gamma^*\Omega^1_{\mathrm{Gr}(s,\Omega^1_X(x))} \to \Omega^1_T$ (Definition 2.7), the restriction of $d\gamma$ to T' factors through the pullback of $\Omega^1_{\operatorname{Spec} k/k} = 0$. Thus $d\gamma = 0$. □

5. Minimal rational curves in characteristic zero

In this section, we construct the morphism κ of Construction-Proposition 5.3 and prove that the relative differential of κ over X is non-zero at some point (Definition-Proposition 5.4). The non-vanishing is an open condition expressing that the non-linearity of the VMRT forces W_g to vary with g when $g(0)$ is a fixed general point, and Section 6 carries the non-vanishing to reductions modulo p, where it contradicts Corollary 4.3.

Set-up 5.1. Let X, d, $\mathcal{K}$, and s be as follows.

(1) X is a Fano manifold of dimension d with $\rho(X) = 1$.
(2) $\mathcal{K}$ is a minimal component of rational curves on X whose VMRT is non-linear.
(3) s is the integer such that the normalization $\nu : \mathbb{P}^1 \to X$ of a general member of $\mathcal{K}$ satisfies

$$\nu^* T_X \cong \mathcal{O}_{\mathbb{P}^1}(2) \oplus \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus(d-1-s)} \oplus \mathcal{O}_{\mathbb{P}^1}^{\oplus s}$$

(see Lemma 2.9).

To write the variation of W_g as a condition that can be spread out, we need an algebraic family of the subspaces W_g over a smooth parameter space, and the definition of M below fixes the splitting type of g^*T_X, so that the dimension of $H^0(\mathbb{P}^1, g^*\Omega^1_X)$ does not depend on g, the subspaces W_g form a subbundle of rank s, and we obtain κ.

Definition-Proposition 5.2. Notation and conditions as in Set-up 5.1. Let $M \subset \mathrm{Hom}(\mathbb{P}^1, X)$ be the set of morphisms g that are birational onto members of $\mathcal{K}$ and satisfy

$$g^* T_X \cong \mathcal{O}_{\mathbb{P}^1}(2) \oplus \mathcal{O}_{\mathbb{P}^1}(1)^{\oplus(d-1-s)} \oplus \mathcal{O}_{\mathbb{P}^1}^{\oplus s},$$

where $\mathrm{Hom}(\mathbb{P}^1, X)$ is the scheme of morphisms from $\mathbb{P}^1$ to X [Kol96, I, Theorem 1.10]. Then M is the set of closed points of an open subscheme of $\mathrm{Hom}(\mathbb{P}^1, X)$, which we also denote by M, and M is non-empty, smooth, and of finite type over $\mathbb{C}$.

Proof. By [Kol96, I, Theorem 1.10] (cf. [AWZ21, Section 6.1]), $\mathrm{Hom}(\mathbb{P}^1, X)$ is an open subscheme of $\mathrm{Hilb}(\mathbb{P}^1 \times X)$, and the morphisms g with $\deg g^*(-K_X) = d + 1 - s$ form an open and closed subscheme $\mathrm{Hom}_{d+1-s}(\mathbb{P}^1, X)$ of $\mathrm{Hom}(\mathbb{P}^1, X)$, which is of finite type over $\mathbb{C}$. Any $g \in M$ satisfies $\deg g^*(-K_X) = \deg g^* T_X = d + 1 - s$, and g is free (Definition-Proposition 2.6) and birational onto $g(\mathbb{P}^1)$. By [Kol96, II, Corollary 3.5.4] (cf. [AWZ21, Section 6.1]), the free morphisms in $\mathrm{Hom}_{d+1-s}(\mathbb{P}^1, X)$ that are generically injective form a smooth open subscheme $\mathrm{Hom}^{\mathrm{free}}_{d+1-s}(\mathbb{P}^1, X)$ of $\mathrm{Hom}(\mathbb{P}^1, X)$, which contains M. In characteristic zero, a morphism $g : \mathbb{P}^1 \to X$ is generically injective if and only if g is birational onto $g(\mathbb{P}^1)$, by generic smoothness [Har77, III, Corollary 10.7] applied to the morphism from $\mathbb{P}^1$ to the normalization of $g(\mathbb{P}^1)$. Any open subscheme of $\mathrm{Hom}^{\mathrm{free}}_{d+1-s}(\mathbb{P}^1, X)$ is smooth and of finite type over $\mathbb{C}$, and M is the set of closed points g of $\mathrm{Hom}^{\mathrm{free}}_{d+1-s}(\mathbb{P}^1, X)$ satisfying the condition that g^*T_X has the splitting type in the definition of M and the condition that $g(\mathbb{P}^1)$ is a member of $\mathcal{K}$. Thus it suffices to show that both conditions are open on $\mathrm{Hom}^{\mathrm{free}}_{d+1-s}(\mathbb{P}^1, X)$.

Let g be a closed point of $\mathrm{Hom}^{\mathrm{free}}_{d+1-s}(\mathbb{P}^1, X)$, and write $g^*T_X \cong \bigoplus_{i=1}^d \mathcal{O}_{\mathbb{P}^1}(a_i)$ with $a_1 \geq \cdots \geq a_d \geq 0$, so that $\sum_{i=1}^d a_i = d + 1 - s$. Since g is an isomorphism over the smooth locus of $g(\mathbb{P}^1)$, the differential $T_{\mathbb{P}^1} \cong \mathcal{O}_{\mathbb{P}^1}(2) \to g^*T_X$ of g is non-zero, so $a_1 \geq 2$. Thus g^*T_X has the splitting type in the definition of M if and only if $a_1 = 2$ and $a_2 \leq 1$, that is,

$$h^0(\mathbb{P}^1, g^*T_X(-3)) = 0 \quad \text{and} \quad h^0(\mathbb{P}^1, g^*T_X(-2)) \leq 1.$$

By [Har77, III, Theorem 12.8], applied to the pullbacks of T_X by the universal morphism $\mathbb{P}^1 \times \mathrm{Hom}^{\mathrm{free}}_{d+1-s}(\mathbb{P}^1, X) \to X$ twisted by $\mathcal{O}_{\mathbb{P}^1}(-3)$ and by $\mathcal{O}_{\mathbb{P}^1}(-2)$, both inequalities define open subsets of $\mathrm{Hom}^{\mathrm{free}}_{d+1-s}(\mathbb{P}^1, X)$.

By [Kol96, II, Theorem 2.15] (cf. [AWZ21, Section 6.1]), the quotient of $\mathrm{Hom}^{\mathrm{free}}_{d+1-s}(\mathbb{P}^1, X)$ by $\mathrm{Aut}(\mathbb{P}^1)$ is a smooth open subscheme $\mathrm{RatCurves}^{\mathrm{free}}_{d+1-s}(X)$ of $\mathrm{RatCurves}^n(X)$, and the quotient morphism sends g to the member $g(\mathbb{P}^1)$. As $\mathrm{RatCurves}^{\mathrm{free}}_{d+1-s}(X)$ is smooth, hence normal, the irreducible components of $\mathrm{RatCurves}^{\mathrm{free}}_{d+1-s}(X)$ are pairwise disjoint [Sta, Tag 033M]. Since $\mathcal{K} \cap \mathrm{RatCurves}^{\mathrm{free}}_{d+1-s}(X)$ is empty or an irreducible component of $\mathrm{RatCurves}^{\mathrm{free}}_{d+1-s}(X)$, it is open in $\mathrm{RatCurves}^{\mathrm{free}}_{d+1-s}(X)$. Thus the condition that $g(\mathbb{P}^1)$ is a member of $\mathcal{K}$ is open on $\mathrm{Hom}^{\mathrm{free}}_{d+1-s}(\mathbb{P}^1, X)$.

Finally, by Lemma 2.9 and Set-up 5.1(3), the normalization of a general member of $\mathcal{K}$ belongs to M, so M is non-empty. □

Construction-Proposition 5.3. Notation and conditions as in Set-up 5.1 and Definition-Proposition 5.2. Let $\pi : \mathbb{P}^1 \times M \to M$ be the projection, $u : \mathbb{P}^1 \times M \to X$ the universal morphism, $e : M \to X$ the restriction of u to $\{0\} \times M = M$, $u_m := u|_{\mathbb{P}^1 \times \{m\}}$ for any point $m \in M$, and $\mathcal{V} := \pi_* u^* \Omega^1_X$. Then the following hold.

(1) e is smooth and dominant.
(2) $\mathcal{V}$ is locally free of rank s, and the natural map $\mathcal{V}(m) \to H^0(\mathbb{P}^1_{k(m)}, u_m^* \Omega^1_X)$ is an isomorphism for any $m \in M$.
(3) The evaluation map $\mathcal{V} \to e^* \Omega^1_X$ at 0 is injective with locally free cokernel.

In particular, $\mathcal{V}$ defines a morphism $\kappa : M \to \mathrm{Gr}_X(s, \Omega^1_X)$ over X such that $\kappa(m) = W_{u_m}$ for any closed point $m \in M$.

Proof. (1) Let $m \in M$ be a closed point and $x := e(m)$. By [Kol96, II, Theorem 1.7], the Zariski tangent space $T_m M$ of M at m is $H^0(\mathbb{P}^1, u_m^* T_X)$, and the Zariski tangent space at m of $e^{-1}(x)$, which parametrizes the morphisms in M sending 0 to x, is $H^0(\mathbb{P}^1, u_m^* T_X(-1))$. Since $e^{-1}(x) = M \times_X \operatorname{Spec} k(x)$, the Zariski tangent space of $e^{-1}(x)$ at m is the kernel of the tangent map $T_m M \to T_x X$ of e [Har77, II, Exercise 2.8]. By [Har77, III, Theorem 5.1],

$$h^0(\mathbb{P}^1, u_m^* T_X) - h^0(\mathbb{P}^1, u_m^* T_X(-1)) = (2d + 1 - s) - (d + 1 - s) = d = \dim T_x X,$$

so the tangent map $T_m M \to T_x X$ is surjective. The connected components of the smooth scheme M are integral. Thus e is smooth by [Har77, III, Proposition 10.4], applied to the restriction of e to any connected component of M. As e is smooth, it is flat, so $e(M)$ is a non-empty open subset of X [Har77, III, Exercise 9.1]. Thus e is dominant.

(2) For any closed point $m \in M$, we have

$$u_m^* \Omega^1_X \cong \mathcal{O}_{\mathbb{P}^1}(-2) \oplus \mathcal{O}_{\mathbb{P}^1}(-1)^{\oplus(d-1-s)} \oplus \mathcal{O}_{\mathbb{P}^1}^{\oplus s},$$

so $h^0(\mathbb{P}^1, u_m^* \Omega^1_X) = s$ [Har77, III, Theorem 5.1]. By [Har77, III, Theorem 12.8], the set of points $m \in M$ with $h^0(\mathbb{P}^1_{k(m)}, u_m^* \Omega^1_X) \geq s + 1$ and the set of points $m \in M$ with $h^0(\mathbb{P}^1_{k(m)}, u_m^* \Omega^1_X) \geq s$ are closed in M. As the closed points of a scheme of finite type over $\mathbb{C}$ are dense [Har77, II, Exercise 3.14], the first set, which contains no closed point of M, is empty, and the second set, which contains all closed points of M, is M. Thus $h^0(\mathbb{P}^1_{k(m)}, u_m^* \Omega^1_X) = s$ for any $m \in M$. Since the connected components of M are integral and $u^* \Omega^1_X$ is locally free, hence flat over M, (2) follows from Grauert's theorem [Har77, III, Corollary 12.9], applied to the restriction of π over any connected component of M.

(3) For any $m \in M$, by (2), the map of fibers $\mathcal{V}(m) \to (e^* \Omega^1_X)(m)$ is the evaluation map $H^0(\mathbb{P}^1_{k(m)}, u_m^* \Omega^1_X) \to (e^* \Omega^1_X)(m)$ at 0. Thus the rank of $\mathcal{V}(m) \to (e^* \Omega^1_X)(m)$ is at most s, and it is equal to s if m is closed (Definition-Proposition 2.6). In local trivializations of $\mathcal{V}$ and $e^* \Omega^1_X$, the points $m \in M$ at which the rank is at most $s - 1$ form the common zero locus of the $s \times s$ minors, which is closed in M and contains no closed point of M, hence is empty [Har77, II, Exercise 3.14]. Thus $\mathcal{V}(m) \to (e^* \Omega^1_X)(m)$ is injective for any $m \in M$, and, as taking fibers is right exact, the cokernel $\mathcal{Q}$ of $\mathcal{V} \to e^* \Omega^1_X$ satisfies $\dim \mathcal{Q}(m) = d - s$ for any $m \in M$. Since M is reduced, $\mathcal{Q}$ is locally free [Har77, II, Exercise 5.8(c)]. Moreover, $\mathcal{V} \to e^* \Omega^1_X$ is injective: in local trivializations, the coordinates of a local section of $\mathcal{V}$ in the kernel are functions vanishing at all points of the reduced scheme M, so they are zero.

By (3) and Definition 2.7, where $1 \leq s \leq d - 2$ (Lemma 2.9), the subsheaf $\mathcal{V} \subset e^* \Omega^1_X$ defines a morphism $\kappa : M \to \mathrm{Gr}_X(s, \Omega^1_X)$ over X. For a closed point $m \in M$ and $x := e(m)$, $\kappa(m)$ is the point of the fiber $\mathrm{Gr}(s, \Omega^1_X(x))$ of $\mathrm{Gr}_X(s, \Omega^1_X)$ over x corresponding to the image of $\mathcal{V}(m) \to \Omega^1_X(x)$, which is W_{u_m} by (2). □

We consider the differential of κ along the fibers of e, that is, the relative differential of κ over X, since Corollary 4.3 concerns only free morphisms sending 0 to a given point. We use the non-vanishing of the relative differential rather than the non-constancy of κ on the fibers of e, since the non-vanishing at a point is an open condition.

Definition-Proposition 5.4. Notation and conditions as in Construction-Proposition 5.3. Let $M^\circ \subset M$ be the set of points $m \in M$ such that $(d_{M/X}\kappa)(m) \neq 0$, where $d_{M/X}\kappa$ is the relative differential of κ over X (Definition 2.7). Then M° is a non-empty open subset of M.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we prove that M° is open in M.

Since e is smooth (Construction-Proposition 5.3(1)), $\Omega^1_{M/X}$ is locally free [Har77, III, Proposition 10.4]. Let $m \in M$ be a point with $(d_{M/X}\kappa)(m) \neq 0$. Then there exist an open neighborhood U of m and a section σ of $\kappa^*\Omega^1_{\mathrm{Gr}_X(s,\Omega^1_X)/X}$ on U such that the image of $d_{M/X}\kappa(\sigma)$ in $\Omega^1_{M/X}(m)$ is non-zero (Definition 2.5). Since $\Omega^1_{M/X}$ is locally free, the points $m' \in U$ such that the image of $d_{M/X}\kappa(\sigma)$ in $\Omega^1_{M/X}(m')$ is non-zero form an open neighborhood of m, which is contained in M°. Thus M° is open.

Step 2. In this step, assuming that M° is empty, we prove that κ is constant on any connected component of $e^{-1}(x)$ for a general point $x \in e(M)$.

Suppose that $M^\circ = \emptyset$. As e is smooth (Construction-Proposition 5.3(1)), it is flat, so $e(M)$ is a non-empty open subset of X [Har77, III, Exercise 9.1]. Let $x \in e(M)$ be a general point. For any closed point $m \in M$, the Zariski tangent space $T_m M$ is $H^0(\mathbb{P}^1, u_m^* T_X)$ [Kol96, II, Theorem 1.7], so $\dim T_m M = 2d + 1 - s$ [Har77, III, Theorem 5.1]. Thus any connected component of M is of dimension $2d+1-s$, and $e^{-1}(x)$ is smooth of dimension $(2d+1-s)-d = d+1-s$ [Har77, III, Propositions 10.1(b) and 10.4]. As $e^{-1}(x)$ is smooth of finite type over $\mathbb{C}$, it has finitely many connected components, and they are integral.

By [Har77, II, Proposition 8.10] and Definition 2.7, the restriction of $d_{M/X}\kappa$ to $e^{-1}(x)$ is the differential $d(\kappa|_{e^{-1}(x)})$ of $\kappa|_{e^{-1}(x)} : e^{-1}(x) \to \mathrm{Gr}(s, \Omega^1_X(x))$. For a closed point $m \in e^{-1}(x)$, by [Har77, II, Proposition 8.7], the map $(d(\kappa|_{e^{-1}(x)}))(m)$ is the transpose of the tangent map $T_m(e^{-1}(x)) \to T_{\kappa(m)} \mathrm{Gr}(s, \Omega^1_X(x))$ of $\kappa|_{e^{-1}(x)}$. Since $M^\circ = \emptyset$, the tangent map $T_m(e^{-1}(x)) \to T_{\kappa(m)} \mathrm{Gr}(s, \Omega^1_X(x))$ is zero for any closed point $m \in e^{-1}(x)$.

Suppose that κ is not constant on a connected component N of $e^{-1}(x)$. Let Z be the closure of $\kappa(N)$ in $\mathrm{Gr}(s, \Omega^1_X(x))$ with the reduced structure. Then Z is integral, and $\dim Z \geq 1$. Let $Z^\circ \subset Z$ be a non-empty open subset which is non-singular [Har77, II, Corollary 8.16]. Then κ restricts to a dominant morphism $N \cap \kappa^{-1}(Z^\circ) \to Z^\circ$ of non-singular varieties. By generic smoothness [Har77, III, Corollary 10.7], there exists a non-empty open subset $Z' \subset Z^\circ$ such that $N \cap \kappa^{-1}(Z') \to Z'$ is smooth. For a closed point $m \in N \cap \kappa^{-1}(Z')$, the tangent map $T_m N \to T_{\kappa(m)} Z'$ is surjective [Har77, III, Proposition 10.4], and $T_{\kappa(m)} Z'$ is a non-zero subspace of $T_{\kappa(m)} \mathrm{Gr}(s, \Omega^1_X(x))$, as $\dim Z' \geq 1$. Thus the tangent map $T_m N \to T_{\kappa(m)} \mathrm{Gr}(s, \Omega^1_X(x))$ is non-zero, a contradiction. Therefore, κ is constant on any connected component of $e^{-1}(x)$.

Step 3. In this step, we find an irreducible component $\mathcal{K}_1$ of $\mathcal{K}_x$ and a connected component M_1 of $e^{-1}(x)$ such that $\dim \mathcal{K}_1 \geq 1$ and the members $u_m(\mathbb{P}^1)$ for closed points $m \in M_1$ form a subset of $\mathcal{K}_1$ containing a non-empty open subset of $\mathcal{K}_1$.

Let $h : M \to \mathcal{K}$ be the morphism sending m to the member $u_m(\mathbb{P}^1)$ [Kol96, II, Theorem 2.15] (cf. [AWZ21, Section 6.1]). Since M is open in $\mathrm{Hom}(\mathbb{P}^1, X)$ and stable under precomposition with automorphisms of $\mathbb{P}^1$, $h(M)$ is open in $\mathcal{K}$ [Kol96, II, Theorem 2.15]. As $x \in e(M)$, $h(M)$ intersects $\mathcal{K}_x$. Let $\mathcal{K}_1$ be an irreducible component of $\mathcal{K}_x$ intersecting $h(M)$. Then $\mathcal{U} := \mathcal{K}_1 \cap h(M)$ is a non-empty open subset of $\mathcal{K}_1$. Let $C \in \mathcal{U}$, and write $C = u_{m'}(\mathbb{P}^1)$ for a closed point $m' \in M$. Then $u_{m'}^{-1}(x) \neq \emptyset$ as $C \in \mathcal{K}_x$, and $u_{m'} \circ \phi \in M$ for an automorphism ϕ of $\mathbb{P}^1$ with $\phi(0) \in u_{m'}^{-1}(x)$. Thus $C = u_m(\mathbb{P}^1)$ for $m := u_{m'} \circ \phi \in e^{-1}(x)$, so $\mathcal{U}$ is contained in the union of the images under h of the finitely many connected components of $e^{-1}(x)$ (**Step 2**). Since $\mathcal{K}_1$ is irreducible and $\mathcal{U}$ is dense in $\mathcal{K}_1$, there exists a connected component M_1 of $e^{-1}(x)$ such that $h(M_1) \cap \mathcal{U}$ is dense in $\mathcal{K}_1$. As $h(M_1)$ is irreducible and contained in $\mathcal{K}_x$, the closure of $h(M_1)$ is $\mathcal{K}_1$. Thus $h(M_1)$ contains a non-empty open subset of $\mathcal{K}_1$ by Chevalley's theorem [Har77, II, Exercise 3.19].

Let $m, m' \in M_1$ be closed points with $h(m) = h(m')$. As u_m and $u_{m'}$ are birational onto the member $u_m(\mathbb{P}^1) = u_{m'}(\mathbb{P}^1)$, $u_{m'} = u_m \circ \phi$ for an automorphism ϕ of $\mathbb{P}^1$ with $\phi(0) \in u_m^{-1}(x)$, and $u_m^{-1}(x)$ is a finite set. Thus the closed points of $M_1 \cap h^{-1}(h(m))$ correspond to automorphisms in finitely many cosets of the stabilizer of 0 in $\mathrm{Aut}(\mathbb{P}^1)$, so $\dim(M_1 \cap h^{-1}(h(m))) \le 2$. Since $\dim M_1 = d + 1 - s \ge 3$ (**Step 2** and Lemma 2.9), $h(M_1)$ is not a point. Thus $\dim \mathcal{K}_1 \ge 1$.

Step 4. In this step, we derive a contradiction from [Hwa09, Proposition 2.1].

Recall that x is a general point of $e(M)$ (**Step 2**). For a general member C of $\mathcal{K}$ and a point y at which C is non-singular, [Hwa09, Section 2] defines $H^0(C, T^*X)_y \subset \Omega^1_X(y)$ as the subspace of the values at the point of $\nu^{-1}(y)$ of the sections in $H^0(\mathbb{P}^1, \nu^*\Omega^1_X)$, where $\nu : \mathbb{P}^1 \to X$ is the normalization of C. We define $H^0(C, T^*X)_y$ in this way for any member C of $\mathcal{K}$ that is non-singular at y. By [Hwa09, Proposition 2.1], if X is a Fano manifold with $\rho(X) = 1$, $\mathcal{K}$ is a minimal component of rational curves on X, $x \in X$ is a general point, $\mathcal{K}_1$ is an irreducible component of $\mathcal{K}_x$, and a non-zero vector $v \in T_xX$ annihilates $H^0(C, T^*X)_x$ for any general member C of $\mathcal{K}_1$, then the VMRT of $\mathcal{K}$ is linear.

By [Keb02, Theorem 3.3], only finitely many members of $\mathcal{K}_x$ are singular at x. Since $\dim \mathcal{K}_1 \ge 1$ and $h(M_1)$ contains a non-empty open subset of $\mathcal{K}_1$ (**Step 3**), a general member C of $\mathcal{K}_1$ is non-singular at x and satisfies $C = u_m(\mathbb{P}^1)$ for a closed point $m \in M_1$. For such C and m, let $\nu : \mathbb{P}^1 \to X$ be the normalization of C. Then $u_m = \nu \circ \phi$ for an automorphism ϕ of $\mathbb{P}^1$. Since $m \in M$, $\nu^*T_X \cong (\phi^{-1})^* u_m^* T_X$ has the splitting type of Set-up 5.1(3). As C is non-singular at x, we have $u_m^{-1}(x) = \{0\}$ and $\nu^{-1}(x) = \{\phi(0)\}$. Since $u_m^*\Omega^1_X = \phi^*\nu^*\Omega^1_X$ and $\nu^{-1}(x) = \{\phi(0)\}$, Construction-Proposition 5.3 and the definition of $H^0(C, T^*X)_x$ imply that

$$\kappa(m) = W_{u_m} = H^0(C, T^*X)_x.$$

By **Step 2**, there exists a subspace $V \subset \Omega^1_X(x)$ such that $\kappa(m) = V$ for any closed point $m \in M_1$. As $\dim V = s < d$, there exists a non-zero vector $v \in T_xX$ annihilating V. Let C be a general member of $\mathcal{K}_1$. Then C is non-singular at x [Keb02, Theorem 3.3], and v annihilates $H^0(C, T^*X)_x = \kappa(m) = V$, where $m \in M_1$ is a closed point with $C = u_m(\mathbb{P}^1)$. Since, moreover, X is a Fano manifold with $\rho(X) = 1$ and $\mathcal{K}$ is a minimal component of rational curves on X (Set-up 5.1(1)(2)), x is a general point of X (**Step 2**), and $\mathcal{K}_1$ is an irreducible component of $\mathcal{K}_x$ (**Step 3**), [Hwa09, Proposition 2.1] implies that the VMRT of $\mathcal{K}$ is linear, where the linearity in [Hwa09, Section 2] is that of Definition 2.8. This contradicts Set-up 5.1(2). Thus M° is non-empty. □

6. Proofs of Theorem 1.2 and Corollary 1.3

In this section, we prove Theorem 1.2 and Corollary 1.3 by reduction modulo a prime, combining Definition-Proposition 5.4, which Construction-Proposition 6.1 carries to the reductions, with Proposition 3.3 and Corollary 4.3.

We spread out X, f, and the family of Construction-Proposition 5.3 over one finitely generated $\mathbb{Z}$-algebra R, so that for any algebraically closed field k and any morphism $t : \mathrm{Spec}\, k \to \mathrm{Spec}\, R$ whose image is a closed point, both the hypotheses of Proposition 3.3 and properties (3) and (4) below hold for t. Property (3) has to hold at the closed points of the reductions of the family along such t, where the splitting type of the curves is not known in advance, and the base change argument that extends it from characteristic zero is carried out over an integral affine open subset of M, so we spread out only an open subset M' of M intersecting M°, which suffices because the proof of Theorem 1.2 uses only one point of the set M_t° in (4).

Construction-Proposition 6.1. Notation and conditions as in Set-up 5.1, Definition-Proposition 5.2, Construction-Proposition 5.3, and Definition-Proposition 5.4. Let $f : X \to X$ be a surjective morphism with $\deg f > 1$. Then there exist an integer $q \ge 2$, an open subset $M' \subset M$ with $M' \cap M^\circ \ne \emptyset$, a finitely generated $\mathbb{Z}$-subalgebra $R \subset \mathbb{C}$ that is smooth over $\mathbb{Z}$ with

$q \in R^\times$, smooth R-schemes X_R and M_R of finite type, and morphisms of R-schemes

$$f_R : X_R \to X_R, \quad u_R : \mathbb{P}^1_R \times_{\operatorname{Spec} R} M_R \to X_R, \quad e_R : M_R \to X_R, \quad \kappa_R : M_R \to \operatorname{Gr}_{X_R}(s, \Omega^1_{X_R/R})$$

such that the base changes of X_R, M_R, f_R, u_R, e_R, and κ_R along $\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} R$ are X, M', f, $u|_{\mathbb{P}^1 \times M'}$, $e|_{M'}$, and $\kappa|_{M'}$, and such that the following hold, where $M_R^\circ \subset M_R$ denotes the set of points at which the relative differential $d_{M_R/X_R}\kappa_R$ is non-zero.

(1) $X_R \to \operatorname{Spec} R$ is projective with geometrically integral fibers, $\omega^{-1}_{X_R/R}$ is ample over $\operatorname{Spec} R$, f_R is finite, and $f_R^*\omega^{-1}_{X_R/R} \cong (\omega^{-1}_{X_R/R})^{\otimes q}$.
(2) e_R is smooth and is the restriction of u_R to $\{0\} \times M_R = M_R$, and κ_R is a morphism over X_R.
(3) For any algebraically closed field k, any morphism $t : \operatorname{Spec} k \to \operatorname{Spec} R$, and any closed point $m \in M_t$, the morphism $u_m := u_t|_{\mathbb{P}^1_k \times \{m\}} : \mathbb{P}^1_k \to X_t$ is free, $h^0(\mathbb{P}^1_k, u_m^*\Omega^1_{X_t}) = s$, and $\kappa_t(m) = W_{u_m}$.
(4) For any algebraically closed field k and any morphism $t : \operatorname{Spec} k \to \operatorname{Spec} R$, the base change M_t° of M_R° is a non-empty open subset of M_t consisting of the points at which $d_{M_t/X_t}\kappa_t$ is non-zero.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step, we construct q, R, X_R, M_R, f_R, u_R, and e_R such that (1) holds and e_R is smooth.

By Lemma 2.11 and $\deg f > 1$, the morphism f is finite, and there exists an integer $q \geq 2$ such that $f^*\omega_X^{-1} \cong (\omega_X^{-1})^{\otimes q}$. Since M is smooth and M° is a non-empty open subset of M (Definition-Proposition 5.4), there exists an integral affine open subset $U \subset M$ with $U \cap M^\circ \neq \emptyset$. By [Gro66, Théorème 8.8.2, Proposition 8.9.1], there exist a finitely generated $\mathbb{Z}$-subalgebra $R \subset \mathbb{C}$, R-schemes X_R and M_R of finite type with M_R affine, and morphisms of R-schemes $f_R : X_R \to X_R$ and $u_R : \mathbb{P}^1_R \times_{\operatorname{Spec} R} M_R \to X_R$, whose base changes along $\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} R$ are X, U, f, and $u|_{\mathbb{P}^1 \times U}$. Let e_R be the composite of u_R with the section $M_R \to \mathbb{P}^1_R \times_{\operatorname{Spec} R} M_R$ at 0, that is, the restriction of u_R to $\{0\} \times M_R = M_R$. Then the base change of e_R is $e|_U$, which is smooth by Construction-Proposition 5.3(1). Since X is smooth and projective, f is finite, and U and $e|_U$ are smooth, possibly replacing R with a larger finitely generated $\mathbb{Z}$-subalgebra of $\mathbb{C}$, we may assume that X_R and M_R are smooth over R, e_R is smooth [Gro67, Proposition 17.7.8], X_R is projective over R, and f_R is finite [Gro66, Théorème 8.10.5]. Since the base change of $\omega_{X_R/R}$ is ω_X [Har77, II, Proposition 8.10] and ω_X^{-1} is ample, possibly enlarging R further, we may assume that $\omega^{-1}_{X_R/R}$ is ample over $\operatorname{Spec} R$ [Gro66, Lemme 8.10.5.2] and $f_R^*\omega^{-1}_{X_R/R} \cong (\omega^{-1}_{X_R/R})^{\otimes q}$ [Gro66, Corollaire 8.5.2.5].

Replacing R with $R[1/a]$ for a non-zero $a \in R$, and the R-schemes and morphisms above with their restrictions over $\operatorname{Spec} R[1/a]$, does not change the base changes along $\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} R$ and keeps the properties above. Since X is integral and $R \subset \mathbb{C}$, by [Gro66, Théorème 12.2.4(viii)] and [Gro67, Théorème 17.5.1, Corollaire 17.15.7], possibly replacing R with such an $R[1/a]$, we may assume that $X_R \to \operatorname{Spec} R$ has geometrically integral fibers, R is smooth over $\mathbb{Z}$, and $q \in R^\times$. Then (1) holds. Since M_R is flat over the domain R, the ring $\mathcal{O}(M_R)$ is torsion-free over R, so the natural map $\mathcal{O}(M_R) \to \mathcal{O}(M_R) \otimes_R \mathbb{C} = \mathcal{O}(U)$ is injective, so M_R is integral. Moreover, for any non-zero $b \in \mathcal{O}(M_R)$, the base change of the principal open subset $D(b) \subset M_R$ is a non-empty open subset of the irreducible U, so it has non-empty intersection with M°.

Step 2. In this step, possibly replacing M_R with a principal open subset, we construct κ_R and prove (2) and (3).

Let $\pi_R : \mathbb{P}^1_R \times_{\operatorname{Spec} R} M_R \to M_R$ be the projection, $\mathcal{E} := u_R^*\Omega^1_{X_R/R}$, $\mathcal{V}_R := \pi_{R*}\mathcal{E}$, and $\mathcal{E}^\vee(-1)$ the tensor product of $u_R^*T_{X_R/R}$ with the pullback of $\mathcal{O}_{\mathbb{P}^1_R}(-1)$, where $T_{X_R/R}$ is the dual of $\Omega^1_{X_R/R}$. For $z \in M_R$ and $\mathcal{G} \in \{\mathcal{E}, \mathcal{E}^\vee(-1)\}$, let $h^i(z, \mathcal{G})$ be the dimension of $H^i(\pi_R^{-1}(z), \mathcal{G}|_{\pi_R^{-1}(z)})$ over $k(z)$.

By [Har77, III, Proposition 12.2], there exists a bounded above complex $L^\bullet$ of finitely generated free $\mathcal{O}(M_R)$-modules, with differentials $\partial^i : L^i \to L^{i+1}$, such that

$$H^i(\mathbb{P}^1_B, \mathcal{E}_B) \cong H^i(L^\bullet \otimes_{\mathcal{O}(M_R)} B)$$

for any $i \geq 0$, functorially in the $\mathcal{O}(M_R)$-algebra B, where $\mathcal{E}_B$ is the pullback of $\mathcal{E}$ to $\mathbb{P}^1_B$, and similarly for $\mathcal{E}^\vee(-1)$. Thus $h^i(z, \mathcal{G})$ does not change under field extensions of $k(z)$.

By Definition-Proposition 5.2 and [Har77, III, Theorem 5.1], for any closed point $m_1 \in U$, we have $h^0(\mathbb{P}^1, u_{m_1}^* \Omega^1_X) = s$ and $H^1(\mathbb{P}^1, u_{m_1}^* T_X \otimes \mathcal{O}_{\mathbb{P}^1}(-1)) = 0$. Thus the image $z \in M_R$ of m_1 satisfies $h^0(z, \mathcal{E}) = s$ and $h^1(z, \mathcal{E}^\vee(-1)) = 0$ [Har77, II, Proposition 8.10]. For $i \in \{-1, 0\}$, let r_i be the rank of ∂^i over the fraction field of the domain $\mathcal{O}(M_R)$, and let $b_0 \neq 0$ be the product of a non-zero $r_{-1} \times r_{-1}$ minor of ∂^{-1} and a non-zero $r_0 \times r_0$ minor of ∂^0. Since the $(r_i + 1) \times (r_i + 1)$ minors of ∂^i are zero, the rank of $\partial^i \otimes k(z)$ is r_i for any $z \in D(b_0)$, so $h^0(z, \mathcal{E})$ does not depend on $z \in D(b_0)$. By [Har77, III, Theorem 12.8], the set of $z \in M_R$ with $h^1(z, \mathcal{E}^\vee(-1)) = 0$ is open, so it contains $D(b_1)$ for some $b_1 \neq 0$, as it contains the image of a closed point of U. Since $D(b_0 b_1)$ is an integral affine open subset of M_R containing the image of a closed point of U (**Step 1**), possibly replacing M_R with $D(b_0 b_1)$, and $L^\bullet$ with $L^\bullet \otimes_{\mathcal{O}(M_R)} \mathcal{O}(D(b_0 b_1))$, we may assume that

$$h^0(z, \mathcal{E}) = s \quad \text{and} \quad h^1(z, \mathcal{E}^\vee(-1)) = 0 \quad \text{for any } z \in M_R.$$

Let $M' \subset U$ be the base change of M_R along $\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} R$. By **Step 1**, $M' \cap M^\circ \neq \emptyset$.

Let k be an algebraically closed field, $t : \operatorname{Spec} k \to \operatorname{Spec} R$ a morphism, $m \in M_t$ a closed point, and $z \in M_R$ the image of m. We identify $(\mathbb{P}^1_R \times_{\operatorname{Spec} R} M_R)_t$ with $\mathbb{P}^1_k \times_{\operatorname{Spec} k} M_t$. Then $u_m^* \Omega^1_{X_t}$ and $u_m^* T_{X_t} \otimes \mathcal{O}_{\mathbb{P}^1_k}(-1)$ are the pullbacks of $\mathcal{E}$ and $\mathcal{E}^\vee(-1)$ along $\mathbb{P}^1_k \to \pi_R^{-1}(z)$ [Har77, II, Proposition 8.10]. Thus $h^0(\mathbb{P}^1_k, u_m^* \Omega^1_{X_t}) = s$. Moreover, writing $u_m^* T_{X_t} \cong \bigoplus_{i=1}^d \mathcal{O}_{\mathbb{P}^1_k}(a_i)$ [Har77, V, Exercise 2.6], the vanishing of $H^1(\mathbb{P}^1_k, u_m^* T_{X_t} \otimes \mathcal{O}_{\mathbb{P}^1_k}(-1))$ implies $a_i \geq 0$ for any i [Har77, III, Theorem 5.1], that is, u_m is free.

Since M_R is integral and $h^0(z, \mathcal{E}) = s$ for any $z \in M_R$, by [Har77, III, Corollary 12.9], the sheaf $\mathcal{V}_R$ is locally free of rank s, and the natural map $\mathcal{V}_R(z) \to H^0(\pi_R^{-1}(z), \mathcal{E}|_{\pi_R^{-1}(z)})$ is an isomorphism for any $z \in M_R$. Let $j_R : \mathcal{V}_R \to e_R^* \Omega^1_{X_R/R}$ be the evaluation map at 0. For any algebraically closed field k, any morphism $t : \operatorname{Spec} k \to \operatorname{Spec} R$, and any closed point $m \in M_t$ with image $z \in M_R$, the map $j_R(z) \otimes_{k(z)} k$ is the evaluation map

$$H^0(\mathbb{P}^1_k, u_m^* \Omega^1_{X_t}) \to \Omega^1_{X_t}(u_m(0))$$

at 0, which is injective with image W_{u_m} by Definition-Proposition 2.6. Since any $z \in M_R$ is the image of a closed point of M_t for k an algebraic closure of $k(z)$ and t the induced morphism, the map $j_R(z)$ is injective for any $z \in M_R$. Thus the cokernel $\mathcal{Q}$ of j_R satisfies $\dim_{k(z)} \mathcal{Q}(z) = d - s$ for any $z \in M_R$. Since M_R is reduced, $\mathcal{Q}$ is locally free [Har77, II, Exercise 5.8(c)], so j_R maps $\mathcal{V}_R$ onto the locally free sheaf $\operatorname{Ker}(e_R^* \Omega^1_{X_R/R} \to \mathcal{Q})$ of rank s, and j_R is injective. By Definition 2.7, the subsheaf $j_R(\mathcal{V}_R) \subset e_R^* \Omega^1_{X_R/R}$ defines a morphism $\kappa_R : M_R \to \operatorname{Gr}_{X_R}(s, \Omega^1_{X_R/R})$ over X_R, where M_R is an X_R-scheme via e_R. Together with **Step 1**, this proves (2).

By [Har77, II, Proposition 8.10] and Definition 2.7, we identify $\operatorname{Gr}_{X_t}(s, \Omega^1_{X_t})$ and $\operatorname{Gr}_X(s, \Omega^1_X)$ with the base changes of $\operatorname{Gr}_{X_R}(s, \Omega^1_{X_R/R})$ along t and along $\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} R$, respectively. Then κ_t corresponds to the pullback of j_R to M_t, so $\kappa_t(m)$ is the image of $j_R(z) \otimes_{k(z)} k$, that is, $\kappa_t(m) = W_{u_m}$. This proves (3). Finally, $\mathcal{O}(M') = \mathcal{O}(M_R) \otimes_R \mathbb{C}$ is flat over $\mathcal{O}(M_R)$, as $\mathbb{C}$ is flat over R. Thus

$$H^0(L^\bullet \otimes_{\mathcal{O}(M_R)} \mathcal{O}(M')) = H^0(L^\bullet) \otimes_{\mathcal{O}(M_R)} \mathcal{O}(M'),$$

so the base changes of $\mathcal{V}_R$ and j_R along $\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} R$ are $\mathcal{V}|_{M'}$ and the evaluation map at 0 of Construction-Proposition 5.3(3). Hence the base change of κ_R is $\kappa|_{M'}$.

Step 3. In this step, possibly replacing R with $R[1/a]$ for a non-zero $a \in R$, we prove (4).

Since e_R is smooth, $\Omega^1_{M_R/X_R}$ is locally free of finite type [Gro67, Proposition 17.2.3(i)]. Let $z \in M^\circ_R$. Then there exists a section σ of

$$\kappa_R^* \Omega^1_{\mathrm{Gr}_{X_R}(s,\Omega^1_{X_R/R})/X_R}$$

over an open neighborhood of z such that the image of $d_{M_R/X_R}\kappa_R(\sigma)$ in $\Omega^1_{M_R/X_R}(z)$ is non-zero (Definition 2.5). Since $\Omega^1_{M_R/X_R}$ is locally free, the points z' at which the image of $d_{M_R/X_R}\kappa_R(\sigma)$ in $\Omega^1_{M_R/X_R}(z')$ is non-zero form an open neighborhood of z contained in M°_R. Thus M°_R is open in M_R. By [Har77, II, Proposition 8.10], for any field K and any morphism $\operatorname{Spec} K \to \operatorname{Spec} R$, the relative differential of the base change of κ_R is the pullback of $d_{M_R/X_R}\kappa_R$. Since taking fibers commutes with pullback, the base change of M°_R is the set of points at which the relative differential of the base change of κ_R is non-zero. In particular, by **Step 2**, the base change of M°_R along $\operatorname{Spec}\mathbb{C} \to \operatorname{Spec} R$ is $M' \cap M^\circ \neq \emptyset$.

Since $M_R \to \operatorname{Spec} R$ is smooth, hence flat of finite type, it is open [Har77, III, Exercise 9.1]. Thus the image of M°_R in $\operatorname{Spec} R$ is an open subset containing the generic point. Replacing R with $R[1/a]$ for a non-zero $a \in R$ keeps (1), (2), (3), and the base changes along $\operatorname{Spec}\mathbb{C} \to \operatorname{Spec} R$. Thus, possibly replacing R with such an $R[1/a]$, we may assume that $M^\circ_R \to \operatorname{Spec} R$ is surjective. Then $M^\circ_t \neq \emptyset$ for any k and t as in (4), and (4) holds. □

Proof of Theorem 1.2. Suppose that $X \ncong \mathbb{P}^d$. By Lemma 2.12, there exists a minimal component $\mathcal{K}$ of rational curves on X whose VMRT is non-linear. By Lemma 2.9, there exists an integer s with $1 \leq s \leq d-2$ (so $d \geq 3$) such that X, d, $\mathcal{K}$, and s satisfy the conditions of Set-up 5.1. Let q, R, X_R, M_R, f_R, u_R, e_R, κ_R, and M°_R be as in Construction-Proposition 6.1.

Let $\mathfrak{m} \subset R$ be a maximal ideal, k an algebraic closure of $R/\mathfrak{m}$, and $t : \operatorname{Spec} k \to \operatorname{Spec} R$ the induced morphism. Since geometrically integral fibers are geometrically connected, Construction-Proposition 6.1 and Construction-Proposition 6.1(1) show that the hypotheses of Proposition 3.3 hold for R, q, X_R, f_R, k, and t. Thus k has positive characteristic, and X_t is F-liftable (Proposition 3.3). Moreover, since X_R is smooth over R (Construction-Proposition 6.1) and projective over R with geometrically integral fibers (Construction-Proposition 6.1(1)), X_t is a smooth projective variety over k [Gro67, Proposition 17.3.3(iii)]. Choose a Frobenius lifting $(\tilde{X}_t, \tilde{F}_{X_t})$ of X_t, and let X°_t be the non-empty open subset of X_t defined in Definition-Lemma 2.3.

By Construction-Proposition 6.1(4), M°_t is a non-empty open subset of M_t. Since e_R is smooth (Construction-Proposition 6.1(2)), the base change e_t of e_R is smooth [Gro67, Proposition 17.3.3(iii)], hence flat of finite type, so e_t is open [Har77, III, Exercise 9.1]. Since X_t is irreducible, $X^\circ_t \cap e_t(M^\circ_t)$ is a non-empty open subset of X_t, so it contains a closed point y [Har77, II, Exercise 3.14]. Choose $m \in M^\circ_t$ with $e_t(m) = y$.

Let $T := e_t^{-1}(y)$, which is smooth over k [Gro67, Proposition 17.3.3(iii)], let $g_T := u_t|_{\mathbb{P}^1_k \times T}$, and for any closed point $\tau \in T$, let $g_\tau := g_T|_{\mathbb{P}^1_k \times \{\tau\}}$. Since κ_t is a morphism over X_t (Construction-Proposition 6.1(2)), the restriction of κ_t to T is a morphism

$$\gamma : T \to \mathrm{Gr}(s, \Omega^1_{X_t}(y))$$

to the fiber of $\mathrm{Gr}_{X_t}(s, \Omega^1_{X_t})$ over y (Definition 2.7). Let $\tau \in T$ be a closed point. Since T is closed in M_t, τ is a closed point of M_t. By Construction-Proposition 6.1(2), e_t is the restriction of u_t to $\{0\} \times M_t$, so $g_\tau(0) = e_t(\tau) = y$. By Construction-Proposition 6.1(3), applied to $u_\tau = g_\tau$, the morphism g_τ is free, $h^0(\mathbb{P}^1_k, g_\tau^* \Omega^1_{X_t}) = s$, and $\gamma(\tau) = \kappa_t(\tau) = W_{g_\tau}$. Since $y \in X^\circ_t$ and T is smooth, Corollary 4.3 implies $d\gamma = 0$.

On the other hand, by [Har77, II, Proposition 8.10], $d\gamma$ is the pullback of $d_{M_t/X_t}\kappa_t$ to T. Thus

$$(d_{M_t/X_t}\kappa_t)(m) = (d\gamma)(m) = 0,$$

which contradicts $m \in M^\circ_t$ (Construction-Proposition 6.1(4)). Hence $X \cong \mathbb{P}^d$. □

Proof of Corollary 1.3. Since X is smooth and rationally connected, by the equivalence of conditions (iii) and (v) in [IP99, Definition-theorem 6.2.1], there exists a morphism $g : \mathbb{P}^1 \to X$ such that $g^*T_X \cong \bigoplus_{i=1}^d \mathcal{O}_{\mathbb{P}^1}(a_i)$ with $a_i > 0$ for any i. Thus

$$K_X \cdot g_*[\mathbb{P}^1] = -\deg g^*T_X = -\sum_{i=1}^d a_i < 0.$$

Since $\rho(X) = 1$ and $-K_X \cdot g_*[\mathbb{P}^1] > 0$, the divisor $-K_X$ is numerically equivalent to a positive multiple of an ample divisor, so $-K_X$ is ample by Kleiman's criterion [KM98, Theorem 1.18]. Thus X is a Fano manifold.

By Lemma 2.11, $\mathrm{Pic}(X)$ is generated by the class of an ample Cartier divisor H, and there exists a positive integer q such that $f^*H \sim qH$ and $\deg f = q^d$. By Definition 2.10, there exists an ample Cartier divisor A on X such that $f^*A - A$ is ample. Write $A \sim aH$ with $a \in \mathbb{Z}$. Since $f^*A - A \sim a(q-1)H$ is ample, we have $a(q-1) > 0$. As q is a positive integer, $q \geq 2$. Thus $\deg f = q^d > 1$. By Theorem 1.2, $X \cong \mathbb{P}^d$. □

References

[AWZ21] P. Achinger, J. Witaszek, and M. Zdanowicz, *Global Frobenius liftability I*, J. Eur. Math. Soc. (JEMS) **23** (2021), no. 8, 2601–2648.

[Ame97] E. Amerik, *Maps onto certain Fano threefolds*, Doc. Math. **2** (1997), 195–211.

[ARV99] E. Amerik, M. Rovinsky, and A. Van de Ven, *A boundedness theorem for morphisms between threefolds*, Ann. Inst. Fourier (Grenoble) **49** (1999), no. 2, 405–415.

[AKP08] M. Aprodu, S. Kebekus, and T. Peternell, *Galois coverings and endomorphisms of projective varieties*, Math. Z. **260** (2008), no. 2, 431–449.

[Bea01] A. Beauville, *Endomorphisms of hypersurfaces and other manifolds*, Internat. Math. Res. Notices (2001), no. 1, 53–58.

[CP91] F. Campana and T. Peternell, *Projective manifolds whose tangent bundles are numerically effective*, Math. Ann. **289** (1991), no. 1, 169–187.

[CMZ26] Z. Chen, S. Meng, and G. Zhong, *On smooth rationally connected projective threefolds of Picard number two admitting int-amplified endomorphisms*, J. Lond. Math. Soc. (2) **114** (2026), no. 1, e70646.

[Cho11] I. Choe, *Endomorphisms of hypersurfaces of Fano manifolds of Picard number 1*, Math. Nachr. **284** (2011), no. 2-3, 213–216.

[Fak03] N. Fakhruddin, *Questions on self maps of algebraic varieties*, J. Ramanujan Math. Soc. **18** (2003), no. 2, 109–122.

[Gro66] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. III*, Inst. Hautes Études Sci. Publ. Math. **28** (1966), 5–255.

[Gro67] A. Grothendieck, *Éléments de géométrie algébrique. IV. Étude locale des schémas et des morphismes de schémas. IV*, Inst. Hautes Études Sci. Publ. Math. **32** (1967), 5–361.

[Har77] R. Hartshorne, *Algebraic geometry*, Grad. Texts in Math. **52**, Springer, New York (1977).

[Hwa07] J.-M. Hwang, *Deformation of holomorphic maps onto Fano manifolds of second and fourth Betti numbers 1*, Ann. Inst. Fourier (Grenoble) **57** (2007), no. 3, 815–823.

[Hwa09] J.-M. Hwang, *Unobstructedness of deformations of holomorphic maps onto Fano manifolds of Picard number 1*, J. Reine Angew. Math. **637** (2009), 193–205.

[HM99] J.-M. Hwang and N. Mok, *Holomorphic maps from rational homogeneous spaces of Picard number 1 onto projective manifolds*, Invent. Math. **136** (1999), no. 1, 209–231.

[HM03] J.-M. Hwang and N. Mok, *Finite morphisms onto Fano manifolds of Picard number 1 which have rational curves with trivial normal bundles*, J. Algebraic Geom. **12** (2003), no. 4, 627–651.

[HM04] J.-M. Hwang and N. Mok, *Birationality of the tangent map for minimal rational curves*, Asian J. Math. **8** (2004), no. 1, 51–64.

[HN11] J.-M. Hwang and N. Nakayama, *On endomorphisms of Fano manifolds of Picard number one*, Pure Appl. Math. Q. **7** (2011), no. 4, 1407–1426.

[HR04] J.-M. Hwang and S. Ramanan, *Hecke curves and Hitchin discriminant*, Ann. Sci. École Norm. Sup. (4) **37** (2004), no. 5, 801–817.

[IP99] V. A. Iskovskikh and Yu. G. Prokhorov, *Fano varieties*, in: *Algebraic geometry V*, Encyclopaedia Math. Sci. **47**, Springer, Berlin (1999), 1–247.

[JZ23] J. Jia and G. Zhong, *Amplified endomorphisms of Fano fourfolds*, Manuscripta Math. **172** (2023), no. 1-2, 567–598.

[JL26] Z. Jiang and Y. Luo, *Varieties admitting polarized Galois endomorphisms*, arXiv:2610.05248.

[KT25] T. Kawakami and B. Totaro, *Endomorphisms of varieties and Bott vanishing*, J. Algebraic Geom. **34** (2025), no. 2, 381–405.
[Keb02] S. Kebekus, *Families of singular rational curves*, J. Algebraic Geom. **11** (2002), no. 2, 245–256.
[Kol96] J. Kollár, *Rational curves on algebraic varieties*, Ergeb. Math. Grenzgeb. (3) **32**, Springer, Berlin (1996).
[KM98] J. Kollár and S. Mori, *Birational geometry of algebraic varieties*, Cambridge Tracts in Math. **134**, Cambridge Univ. Press, Cambridge (1998).
[Laz84] R. Lazarsfeld, *Some applications of the theory of positive vector bundles*, in: *Complete intersections (Acireale, 1983)*, Lecture Notes in Math. **1092**, Springer, Berlin (1984), 29–61.
[Laz04] R. Lazarsfeld, *Positivity in algebraic geometry. I. Classical setting: line bundles and linear series*, Ergeb. Math. Grenzgeb. (3) **48**, Springer, Berlin (2004).
[MZ26] S. Matsumura and G. Zhong, *Positivity of compact Kähler varieties admitting an int-amplified endomorphism*, arXiv:2609.09869.
[Men20] S. Meng, *Building blocks of amplified endomorphisms of normal projective varieties*, Math. Z. **294** (2020), no. 3-4, 1727–1747.
[MZZ22] S. Meng, D.-Q. Zhang, and G. Zhong, *Non-isomorphic endomorphisms of Fano threefolds*, Math. Ann. **383** (2022), no. 3-4, 1567–1596.
[MZ23] S. Meng and G. Zhong, *Rigidity of rationally connected smooth projective varieties from dynamical viewpoints*, Math. Res. Lett. **30** (2023), no. 2, 589–610.
[Nak02] N. Nakayama, *Ruled surfaces with non-trivial surjective endomorphisms*, Kyushu J. Math. **56** (2002), no. 2, 433–446.
[PS89] K. H. Paranjape and V. Srinivas, *Self maps of homogeneous spaces*, Invent. Math. **98** (1989), no. 2, 425–444.
[SZ24] F. Shao and G. Zhong, *Boundedness of finite morphisms onto Fano manifolds with large Fano index*, J. Algebra **639** (2024), 678–707.
[SZ25] F. Shao and G. Zhong, *Bigness of tangent bundles and dynamical rigidity of Fano manifolds of Picard number 1* (with an appendix by Jie Liu), Math. Ann. **391** (2025), no. 2, 1731–1752.
[Sta] The Stacks Project Authors, *The Stacks Project*, https://stacks.math.columbia.edu, 2026.
[Wan26] J. Wang, *Families of smooth Fano fourfolds of Picard rank 1 without Bott vanishing*, arXiv:2606.13466.
[Yos21] S. Yoshikawa, *Structure of Fano fibrations of varieties admitting an int-amplified endomorphism*, Adv. Math. **391** (2021), Paper No. 107964.
[Zda18] M. Zdanowicz, *Liftability of singularities and their Frobenius morphism modulo p^2*, Int. Math. Res. Not. IMRN **2018**, no. 14, 4513–4577.

School of Mathematical Sciences, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Beijing International Center for Mathematical Research, Peking University, No. 5 Yiheyuan Road, Haidian District, Beijing 100871, China

Email address: liujihao@math.pku.edu.cn